

Reward-Driven Learning under Prompt-Level Differential Privacy

Jiachen Zhao
University of Notre Dame
jzhao7@nd.edu

Antonia Januszewicz
University of Notre Dame
ajanusze@nd.edu

Taeho Jung*
University of Notre Dame
tjung@nd.edu

Abstract—Reinforcement learning with verifiable rewards (RLVR) trains a language model on problems that may themselves be confidential, and the trained model can reveal which problems it saw. We study RLVR under prompt-level differential privacy: the released weights must be (ϵ, δ) -differentially private with respect to the presence of any one training problem. Taking the group of responses to one prompt as the privacy record, our method aggregates their gradients, clips the prompt’s contribution once, adds Gaussian noise, and composes the privacy loss across updates, so the budget depends on neither the number of responses per prompt nor the clipping norm; to our knowledge this is the first differential privacy guarantee for RLVR training. We train Qwen2.5-1.5B-Instruct with LoRA at a per-run budget of $\epsilon = 8$ and compare, on the same prompts and at the same budget, a control that removes only the reward signal and two private supervised fine-tuning recipes. The reward signal improves accuracy over the control by 2.65 points on MATH and 3.24 on GSM8K, in every seed; the improvement survives a format-robust scorer, at 1.3 points on MATH, and is not explained by response length. At the same budget the private model outperforms both supervised recipes on MATH and GSM8K by 2.3 to 3.8 points, retains 85–90% of the gain of non-private GRPO on these tasks, and on MATH the noise of an eightfold tighter budget costs at most 1.2 points. The reward effect also carries to CommonsenseQA, an exploratory non-mathematical task. Verifier feedback thus remains a usable learning signal under prompt-level privacy.

Index Terms—differential privacy, reinforcement learning with verifiable rewards, group relative policy optimization

I. INTRODUCTION

Reinforcement learning with verifiable rewards (RLVR) improves a language model from its own responses: for each training problem the policy samples several candidate solutions, a program checks each against the problem’s reference answer, and the policy is updated toward the ones that pass [1], [2]. Needing neither demonstrations nor a learned reward model, the method fits settings where solution traces are scarce and correctness is cheap to check: assessment items with answer keys, cases with adjudicated outcomes, tickets with confirmed resolutions. In such settings the training problems are themselves confidential, and RLVR reads each one intensively. Every time a problem is sampled, the policy conditions G generations on it, every reward is computed against its reference answer, and the resulting update is written into the released weights. Membership auditing already detects prompt-specific traces in RLVR-trained models [3]. A model trained on

confidential problems therefore needs a guarantee about what its weights reveal about them, and to our knowledge none exists for RLVR.

Two features of RLVR fix where that guarantee must live and for what unit. In RLHF the sensitive data are human preferences, so private RLHF trains the reward model under DP [4], [5] and the policy optimized against it is covered by post-processing. RLVR has no reward model to privatize: the verifier is public code, and the sensitive object is the prompt the policy reads at every rollout. The gradient step is the only stage that touches private data, so it must be the private mechanism, as in DP-SGD [6] and private PPO [7]. The unit is the prompt: a training record is a problem together with the reference answer its verifier checks, the data are defined at no finer granularity, and presence of one prompt is the event the membership attack recovers. Protection at this unit composes upward; when one person contributes several prompts, a guarantee for that person follows by group privacy or by accounting at the user boundary [8]. We therefore require *prompt-level* differential privacy: the released weights must be (ϵ, δ) -DP with respect to adding or removing any one training problem (Definition 3). This is a guarantee about training items; a guarantee for the people who contribute them is derived from it, at a cost that grows with the number of items per person (Section IV).

The obstacle is the group structure of group relative policy optimization (GRPO) [2]. DP-SGD protects the unit whose gradient is clipped [6], and the direct extension of private policy optimization clips each sampled response, as private PPO does with its one response per prompt [7]. GRPO’s G completions per prompt are not independent contributions: their rewards are normalized within the group, so changing one reward changes every advantage, and the loss averages over the group. Removing a prompt removes G clipped terms, so a per-completion mechanism has sensitivity GC rather than C , and protecting the prompt through it costs group privacy over G records (Proposition 1). User-level DP meets the same problem when one user contributes several examples and resolves it by clipping the user’s aggregated contribution [8], [9]. We do the same for the prompt group: the losses of all G completions are aggregated into one prompt gradient, which is clipped once and perturbed once. Because GRPO’s normalization uses only the group’s own rewards, this gradient depends on the prompt alone given the current parameters, so the sensitivity is C for

*Corresponding author.

every G and the budget depends only on the sampling rate, the noise multiplier, the number of rounds, and δ . Group size and clipping norm become utility and compute parameters with no privacy cost. DP-GRPO is DP-SGD with this record and the single-update GRPO loss as the per-record objective, applied to LoRA parameters [10], [11], and Theorem 1 gives its guarantee by standard subsampled-Gaussian accounting.

Whether anything is learned under this mechanism is the open question, and the answer is not obvious. Each prompt’s influence is clipped to a norm that a single noisy step can dominate, and the learning signal is a within-group contrast whose sign carries the information. Answering it requires isolating the reward signal from everything else the private pipeline does, which existing evaluations of private post-training do not attempt. We train one 1.5B-parameter instruction model with rank-4 LoRA under a fixed protocol: a design study on MATH sets the operating point, and the same arms then run on MATH, on GSM8K as the confirmatory benchmark, and on CommonsenseQA as an exploratory non-mathematical task, every arm spending the same per-run budget, $\epsilon = 8$, on the same prompts. A zero-advantage control, the same private pipeline with every group-relative advantage set to zero after generation and verification, isolates the reward signal. Two private imitation arms, trained on verifier-filtered and on reference solutions under the same sampling, clipping, noise, and accountant, test the other way to spend that budget. Endpoints are fixed in advance, null results carry minimum detectable effects, and every gain is checked against a format-robust scorer and against response length. Three findings result.

(C1) *Verifier rewards improve the policy under prompt-level privacy.* DP-GRPO exceeds the zero-advantage control by +2.65 [1.79, 3.51] pp on MATH and +3.24 [1.61, 4.89] pp on GSM8K, with every seed-level contrast positive, while the control stays within 0.4 points of the base, as expected once the reward term is removed. The contrast survives a format-robust scorer that also credits correct unboxed answers, staying resolved at +1.34 [0.54, 2.14] pp on MATH and positive in every seed, and it does not come from shorter responses: DP-GRPO’s responses are as long as the base’s or longer. The effect carries beyond mathematics: on CommonsenseQA, DP-GRPO exceeds the control by +4.26 [2.50, 6.06] pp under a final-answer rescore.

(C2) *At a matched budget, reward optimization outperforms private imitation on the mathematical tasks.* DP-GRPO exceeds verifier-filtered and gold DP-SFT by 2.32 to 3.79 pp on MATH and GSM8K, and exceeds MATH variants trained at its learning rate or for twice as many updates by 2.10 and 1.78 pp. Neither supervised arm moves beyond its detectable range of the base, even though gold DP-SFT trains on the worked reference solutions as token-level targets, which DP-GRPO touches only through the verifier’s answer check. The comparison is between the evaluated recipes at a matched budget, and the same supervised pipeline does learn where its targets are short: on CommonsenseQA, whose gold targets are four-token answer declarations, gold DP-SFT ties DP-GRPO.

(C3) *At this operating point, the price of privacy is small on the mathematical tasks and the budget can be tightened without a measured loss.* Against non-private GRPO runs measured over their own bases, DP-GRPO retains 85% of the gain on MATH and 90% on GSM8K. On MATH, the noise of an eightfold tighter budget, $\epsilon=1$, changes accuracy by -0.57 [-1.19 , 0.06] pp, a cost of at most 1.2 points (a utility measurement of the procedure at that noise level, Section VII-B), and no tested clipping norm changes it beyond its MDE. Where more remains to learn the price rises: on CommonsenseQA DP-GRPO retains 45% of the non-private gain, and a non-private run at the same hyperparameters is 5.8 points higher. Section IX traces this pattern to the ratio of noise to clipped signal, which shortens each step under AdamW and costs most where the attainable gain is largest; on the mathematical tasks that gain is bounded by the 1.5B-parameter base itself.

This paper makes three contributions.

- *A prompt-level privacy guarantee for RLVR.* Taking the prompt group as the record and clipping its aggregated gradient makes the released LoRA trajectory (ϵ, δ) -DP with respect to any one training problem under stated assumptions (Section IV), with a budget independent of the group size and the clipping norm (Algorithm 1, Lemma 1, Theorem 1). To our knowledge this is the first differential privacy guarantee for GRPO or RLVR training. The same accounting covers the private supervised baselines (Corollary 2), so every comparison in the paper is at a matched guarantee, and mechanism-verification tests check the implementation invariants the analysis relies on: cross-prompt isolation, the clipping identity, the sensitivity bound and Poisson sampling (Appendix B).
- *A protocol that isolates what the reward signal contributes under privacy.* The zero-advantage control removes only the advantages from an otherwise identical private pipeline; matched budgets, fixed endpoints, minimum detectable effects for null results, format-robust rescoring and response-length checks separate reward learning from private training as such, from answer formatting, and from response length. The protocol applies to any private post-training method.
- *Evidence that reward learning survives prompt-level privacy, and where privacy costs.* On two mathematical-reasoning benchmarks and an exploratory non-mathematical task, verifier-derived group-relative rewards remain a learning signal under the mechanism, outperform the evaluated private imitation recipes on the mathematical tasks, retain most of the non-private gain there, and are insensitive to the budget down to $\epsilon=1$ on MATH; the larger cost on CommonsenseQA locates where the noise binds.

II. RELATED WORK

A. Which stage of post-training is private

Private reward-based post-training differs mainly in which stage reads private data and therefore carries the guarantee.

When the sensitive data are human preferences, one route privatizes the reward model, at the user level [4] or through decoupled reward modeling [5], and the policy trained against it is covered by post-processing. A second route privatizes the policy update itself: private PPO on LoRA parameters [7], DP-AdamW applied to the DPO and PPO stages [12], and trajectory-level DP policy gradient [13], in each case with a learned reward and one response per record. A third route avoids training on the private data at all: it fine-tunes a generator under DP, samples synthetic instructions, and runs ordinary alignment on those [14]. RLVR closes off the first and third routes. There is no reward model to privatize, and a synthetic problem comes without the verified reference answer that the reward requires; the prompt is what the policy reads at every rollout (Section I), so the guarantee must attach to the policy update, as in the second route, but for a record that GRPO’s group structure defines differently (Section III-D). The leakage this addresses is documented: membership auditing detects prompt-specific traces in RLVR-trained models [3], and GRPO on benign factual data amplifies extraction of memorized personal data [15]. Neither work evaluates a defense. Adjacent lines answer different questions. KL-regularized sampling makes a sampled action private [16], not a released weight trajectory; private RL theory gives sample-complexity and regret bounds [17], [18] without a mechanism for LLM post-training; DP-OPD trains a student with DP-SGD on its own generations but scores them with a frozen teacher [19]; and RL used *for* privacy, tuning DP-SGD [20], steering a DP text generator [21] or training anonymizers [22], does not protect the training prompts. To our knowledge, no prior work states a differential privacy guarantee for GRPO or RLVR training.

B. Closest work

Wu et al. [7] generate responses to private inputs, train the policy with LoRA and a DP variant of PPO, and release the policy. We keep that release object and differ in three ways. The learning signal is GRPO’s group-relative advantage, which couples every response to its prompt and changes both the record and the clipping unit. The reward is a public verifier, which removes the reward model from the privacy analysis and from the budget. And the evaluation includes a matched intervention that removes only the advantages, so the contribution of the reward signal inside the private procedure is measured rather than assumed.

C. The privacy record

DP-SGD protects whatever is clipped [6]. User-level DP enlarges the record to all of one user’s examples and shows that the enlargement changes both the guarantee and the best training recipe [8], [9]. GRPO poses the question from the other side: the responses to one prompt are generated from it and normalized against each other, and the GRPO gradient is a U-statistic over the group [23], so per-response clipping does not bound the prompt’s influence. Our record is the prompt with all of its responses, and it differs from a user-level record in kind, not only in name: a user’s examples are fixed data, whereas a

prompt’s responses are drawn afresh from the current policy in every round, so the record has no fixed members, no examples-per-unit parameter arises, and the guarantee must hold over the randomness of generation, which is what assumptions (A2) and (A3) of Section VI secure. With the record fixed, low-rank adaptation [11], scale and optimizer choices [24], [25], and privacy-loss-distribution accounting [26]–[28] carry over unchanged.

D. What a credible gain requires

On the privacy side, membership attacks [3] demonstrate leakage and one-run auditing [29] lower-bounds it; neither upper-bounds it, so neither replaces the mechanism proof, and we report our canary diagnostic on those terms (Appendix B). On the utility side, an accuracy gain after RLVR is easy to misattribute. The large gains reported for small Qwen models start from base rather than instruction-tuned checkpoints [30]; part of any gain is format correction [31]; RLVR mainly reweights solutions the base already samples [32]; Qwen2.5-Math improves even under random rewards [33], which has been attributed to contamination [34]; and small gains move with seed, decoding, answer extraction [35] and GRPO’s length bias [36]. These findings set the protocol of Section VII: an already post-trained starting model [37], a control that holds generation, verification, clipping and noise fixed and removes only the advantages, fixed endpoints across seeds, and format-robust and length checks.

E. Private supervised alternatives

Supervised fine-tuning is the other way to spend a prompt-level budget: raise the likelihood of a target response instead of weighting sampled responses by their rewards, under DP-SGD with parameter-efficient adaptation [6], [11], [24]. Our two comparators (Section VII) differ in where the targets come from. Verifier-filtered DP-SFT descends from rejection-sampling fine-tuning [38] and the self-training loops of STaR [39] and ReST^{EM} [40], applied for one round under the same prompt-level mechanism; gold DP-SFT imitates the dataset’s reference solutions. DeepSeekMath places supervised, rejection-sampling and policy-gradient updates in one framework [2], and their empirical ranking is setting-dependent: self-training can beat imitation of human solutions [40], and RL-versus-SFT comparisons turn on the targets and the evaluation distribution [35], [41], [42]. Our comparison holds the starting model, prompt set, clipping mechanism and per-run budget fixed and, because the supervised clipping norm and learning rates differ from GRPO’s, compares specified private procedures rather than objectives in the abstract.

III. BACKGROUND AND PRELIMINARIES

This section fixes the objects the rest of the paper uses: the differential privacy definition and the two facts about it we need (Section III-A), the GRPO objective (Section III-B), the DP-SGD mechanism and its privacy profile (Section III-C), and, from these, the privacy record for GRPO and the guarantee that Theorem 1 proves (Section III-D).

A. Differential Privacy

A dataset D is a finite multiset of records from a universe $\mathcal{X}$. Two datasets D and D' are *neighbors*, written $D \simeq D'$, if one is obtained from the other by adding or removing exactly one record; Definition 3 specializes $\mathcal{X}$ and this relation to prompts.

Definition 1 (Differential privacy [43]): A randomized mechanism $\mathcal{M}$ is (ϵ, δ) -differentially private if, for all $D \simeq D'$ and every measurable set O of outputs,

$$\Pr[\mathcal{M}(D) \in O] \leq e^\epsilon \Pr[\mathcal{M}(D') \in O] + \delta. \quad (1)$$

Proposition 1 (Group privacy): If $\mathcal{M}$ is (ϵ, δ) -DP and D, D' differ in k records, then for every measurable O , $\Pr[\mathcal{M}(D) \in O] \leq e^{k\epsilon} \Pr[\mathcal{M}(D') \in O] + k e^{(k-1)\epsilon} \delta$.

Chaining Definition 1 over k add-or-remove steps gives the additive term $\delta \sum_{j < k} e^{j\epsilon}$; compare [44, Lemma 2.2].

Proposition 2 (Post-processing [43]): If $\mathcal{M}$ is (ϵ, δ) -DP and f is any randomized map that does not depend on D , then $f \circ \mathcal{M}$ is (ϵ, δ) -DP.

B. RLVR and GRPO

A policy π_θ is an autoregressive language model: given a prompt x it defines a distribution over completions $y = (y_1, \dots, y_L)$ through $\pi_\theta(y | x) = \prod_{k=1}^L \pi_\theta(y_k | x, y_{<k})$. In reinforcement learning with verifiable rewards (RLVR) [1], each prompt carries a reference answer, and a programmatic verifier $r(x, y) \in \{0, 1\}$ reports whether the final answer in y matches it. Training maximizes the expected reward $\mathbb{E}_x \mathbb{E}_{y \sim \pi_\theta(\cdot | x)}[r(x, y)]$ under a penalty of weight β on the divergence of π_θ from a fixed reference policy π_{ref} , here the public base model.

Group relative policy optimization (GRPO) [2] estimates this objective without a value function. For prompt x_i it draws G completions $y_{i,1}, \dots, y_{i,G}$ independently from a sampling policy π_{old} , computes rewards $r_{i,g} = r(x_i, y_{i,g})$, and forms group-normalized advantages

$$A_{i,g} = \frac{r_{i,g} - \mu_i}{s_i + c}, \quad (2)$$

where μ_i and s_i are the mean and standard deviation of $\{r_{i,g}\}_{g=1}^G$ and $c > 0$ is a stabilizing constant. Every token of $y_{i,g}$ receives the advantage $A_{i,g}$. With the per-token probability ratio $\rho_{i,g,k} = \pi_\theta(y_{i,g,k} | x_i, y_{i,g,<k}) / \pi_{\text{old}}(y_{i,g,k} | x_i, y_{i,g,<k})$ and completion length $L_{i,g}$, the objective for prompt x_i is [2, [45]

$$J_i(\theta) = \frac{1}{G} \sum_{g=1}^G \frac{1}{L_{i,g}} \sum_{k=1}^{L_{i,g}} \left[\min(\rho_{i,g,k} A_{i,g}, \text{clip}(\rho_{i,g,k}, 1 - \kappa, 1 + \kappa) A_{i,g}) - \beta K_{i,g,k} \right], \quad (3)$$

where $K_{i,g,k} = \pi_{\text{ref}} / \pi_\theta - \log(\pi_{\text{ref}} / \pi_\theta) - 1$, evaluated at token $y_{i,g,k}$, is the nonnegative per-token KL estimator of Schulman [46] adopted by DeepSeekMath [2]. Under samples from π_θ it is an unbiased estimate of the token-level KL($\pi_\theta \| \pi_{\text{ref}}$); its gradient is not an unbiased estimate of the gradient of that divergence [47]. We use it as implemented,

since the privacy analysis of Section VI holds for any per-prompt loss. When one gradient step is taken per batch of rollouts, as in this paper, π_{old} and π_θ coincide at the point of differentiation, so $\rho_{i,g,k} = 1$, the ratio clip is inactive, and $\nabla_\theta \rho_{i,g,k} = \nabla_\theta \log \pi_\theta(y_{i,g,k} | x_i, y_{i,g,<k})$.

C. DP-SGD and the Subsampled Gaussian Mechanism

DP-SGD [6] trains parameters θ on a dataset of n records with per-record losses $\ell_i(\theta)$. Round t performs four steps. (i) *Poisson sampling*: each record i is included in S_t independently with probability q . (ii) *Per-record clipping*: for $i \in S_t$, the gradient $g_i = \nabla_\theta \ell_i(\theta_t)$ is projected to ℓ_2 norm at most C ,

$$\bar{g}_i = g_i \min\left(1, \frac{C}{\|g_i\|_2}\right). \quad (4)$$

(iii) *Noisy aggregation*: the round releases $\hat{g}_t = (\sum_{i \in S_t} \bar{g}_i + z_t) / (qn)$ with $z_t \sim \mathcal{N}(0, \sigma^2 C^2 I)$, where σ is the noise multiplier and qn is the public expected sample size. (iv) *Update*: θ_{t+1} is computed from θ_t and $\hat{g}_t$ by a public optimizer.

Adding or removing one record changes the sum inside (iii) by a vector of norm at most C . Rotating that vector onto a coordinate axis and dividing by C reduces one round, for fixed θ_t , to a pair of one-dimensional output laws, and T adaptively composed rounds to their T -fold products [48], [49], whose hockey-stick divergence is the privacy profile [48].

Definition 2 (Privacy profile of DP-SGD): For probability measures P, Q and $\alpha \geq 0$, let $H_\alpha(P \| Q) = \sup_O [P(O) - \alpha Q(O)]$. For $q \in (0, 1]$ and $\sigma > 0$, let

$$A_{q,\sigma} = (1 - q) \mathcal{N}(0, \sigma^2) + q \mathcal{N}(1, \sigma^2), \quad B_\sigma = \mathcal{N}(0, \sigma^2). \quad (5)$$

The T -round privacy profile is, for $\epsilon \geq 0$,

$$\delta_{q,\sigma,T}(\epsilon) = \max\{H_{e^\epsilon}(A_{q,\sigma}^{\otimes T} \| B_\sigma^{\otimes T}), H_{e^\epsilon}(B_\sigma^{\otimes T} \| A_{q,\sigma}^{\otimes T})\}. \quad (6)$$

Since $H_{e^\epsilon}(\mathcal{M}(D) \| \mathcal{M}(D'))$ is the smallest δ for which (1) holds on the ordered pair (D, D') [48], T rounds of DP-SGD are $(\epsilon, \delta_{q,\sigma,T}(\epsilon))$ -DP for every $\epsilon \geq 0$, and the bound is attained in the worst case over losses [49]. The two terms in (6) are the remove and add directions. The profile depends on (q, σ, T) and not on C , because sensitivity and noise both scale with C . Accountants evaluate it through Rényi divergences [50], [51] or through the privacy loss distribution [26], [27]; Section VI-C states which one this paper reports.

Two remarks on the mechanism are used below. First, the sensitivity bound requires that, given θ_t , the clipped vector $\bar{g}_i$ be a function of record i alone; a statistic shared across records before clipping, such as a batch-level normalization, would let one record change another's contribution and void the bound. Second, the clipping norm C in (4) is unrelated to the ratio clip κ in (3); only C enters the privacy analysis.

a) *Low-rank adaptation*: LoRA [10] freezes a pretrained weight matrix $W \in \mathbb{R}^{d \times k}$ and trains an update $W + BA$ with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. Under DP-SGD only the adapter parameters are clipped and perturbed, and the frozen base weights are public. With p trainable coordinates the noise z_t has expected squared norm $\sigma^2 C^2 p$, so restricting

p lowers the noise at a fixed budget, which is the basis of parameter-efficient private fine-tuning [11], [52].

D. The Privacy Record in GRPO

DP-SGD protects the unit whose gradient is clipped in (4), and a unit spanning k such records is protected only through Proposition 1. The record is therefore a design choice. User-level DP enlarges it from one example to the aggregated contribution of all examples of one user, which is clipped as a whole [8], [9]; there the members of a record are fixed data. Private PPO takes the prompt as the record but generates only one response per prompt [7]. GRPO's record has neither property: its members are G responses sampled from the current policy, so they change every round and are coupled to one another through the group statistics.

In GRPO the natural per-example unit, one completion $y_{i,g}$, is not an independent contribution. The G completions of prompt x_i share the input from which they are sampled and against which they are verified, share the statistics μ_i and s_i of (2), so that changing one reward changes every advantage of the group, and share the average in (3). If each completion were a record clipped to C , removing x_i would remove G clipped terms from the sum in step (iii) of Section III-C, giving sensitivity GC ; a guarantee for the prompt would then follow only from Proposition 1 with $k = G$, or from noise scaled to GC .

We therefore take the record to be the prompt with all of its completions: the gradient g_i of the per-prompt loss $-J_i(\theta_t)$, which already aggregates the G completions, is the quantity clipped in (4). Its sensitivity is C for every G , so the group size does not enter the privacy budget. The first remark of Section III-C holds because μ_i and s_i are computed within the group of x_i ; batch-level reward normalization or advantage whitening across prompts would break it. A record is accordingly a problem together with the reference answer its verifier checks, and we write $x_i \in \mathcal{X}$ for the pair. Because Poisson sampling runs over a fixed index set, we state the guarantee over a public universe of n slots, each holding at most one record.

Definition 3 (Prompt-level differential privacy): Fix a public n . A *prompt dataset* is $D = ((b_i, x_i))_{i=1}^n$ with presence bits $b_i \in \{0, 1\}$ and records $x_i \in \mathcal{X}$, where x_i is read only if $b_i = 1$. Prompt datasets D and D' are neighbors, $D \simeq D'$, if $b_k \neq b'_k$ for exactly one slot k and $(b_i, x_i) = (b'_i, x'_i)$ for every $i \neq k$; the record at slot k on the side where $b_k = 0$ is never read and may be arbitrary. A randomized training algorithm $\mathcal{A}$ with a public round count T that maps a prompt dataset to a weight trajectory $(\theta_2, \dots, \theta_{T+1})$ is *prompt-level* (ϵ, δ) -DP if (1) holds with $\mathcal{M} = \mathcal{A}$ for every pair of neighboring prompt datasets and every measurable set O of trajectories.

Definition 3 is Definition 1 with the record fixed to a prompt group and the output fixed to the whole released trajectory. Section IV states what the adversary sees, Section V gives the algorithm, and Theorem 1 proves that it is prompt-level $(\epsilon, \delta_{q,\sigma,T}(\epsilon))$ -DP.

IV. THREAT MODEL

a) Setting.: We adopt the central model with a trusted curator, the convention of DP-SGD [6], [8]. The curator holds a prompt dataset D over n public slots (Definition 3), with n chosen without access to the data, and executes Algorithm 1; an absent slot contributes the zero vector to every sum. The release is the full trajectory of LoRA weights $\theta_2, \dots, \theta_{T+1}$ together with any checkpoint derived from it. Neighbors keep the capacity n and differ in one presence bit: the standard DP-SGD relation with the record enlarged from one example to one prompt group, under which Poisson sampling over $[n]$ reveals nothing about the realized dataset size.

b) Protected event.: The protected event is the presence of one prompt in D , the value of a single bit b_k . Everything that prompt induces is protected with it: the G rollouts sampled from it in every round in which it is drawn, their rewards and advantages, and its clipped gradient. The output is the whole trajectory because a present prompt is resampled independently in every round, so protecting its presence in one batch would not bound what the trajectory reveals. Membership is the right event for RLVR because it is what an adversary already recovers from an unprotected model: the policy reads each prompt at every rollout, and membership auditing detects prompt-specific traces in RLVR-trained models [3]. Concretely, prompt-level (ϵ, δ) -DP bounds every test of b_k that the adversary below can run on the release: its true-positive rate is at most e^ϵ times its false-positive rate plus δ [53], [54]. We report $\epsilon=8$, the operating point at which private fine-tuning of language models is commonly reported [11], [24]; the budget sweep of Section VII-B measures what the noise of a tighter budget costs.

c) Adversary.: The adversary is white-box, adaptive, and computationally unbounded. It knows every public input of Algorithm 1: the base model, the LoRA initialization θ_1 , the reference policy, the verifier, the optimizer, the hyperparameters $(q, C, \sigma, T, G, \tau, \beta)$, the slot count n , and the accountant. It knows every prompt in D except the one whose presence is in question, may inspect and run the weights, issue unlimited queries, and combine the results with arbitrary side information; each such use is post-processing (Proposition 2) and leaves the guarantee unchanged. It does not observe the run's internal randomness: the sampling coins of line 2, the rollout randomness of line 4, and the Gaussian noise of line 8. Recovery files that store these generator states are not part of the release, and publishing them would void the guarantee.

d) Limits of the guarantee.: Five limits remain, and each is a risk the proof does not remove. (i) *The unit is a prompt, not a person.* If one person contributes k prompts, that person's guarantee follows only through group privacy (Proposition 1) and degrades with k ; protecting people requires grouping and accounting at the user boundary [8]. (ii) *The theorem certifies the analyzed mechanism under assumptions (A1)–(A4) of Section VI.* As in the private fine-tuning literature [11], [24], the budget covers the reported run and not the pilot runs on which its hyperparameters were selected; charging those

Algorithm 1 DP-GRPO: GRPO under prompt-level DP**Input:** Private prompt dataset $D = ((b_i, x_i))_{i=1}^n$ (Definition 3).Public: LoRA initialization θ_1 , reference policy π_{ref} , verifier r , group size $G \geq 2$, sampling temperature τ , advantage constant c , sampling rate q , clipping norm C , noise multiplier σ , rounds T , KL weight β , optimizer Opt.**Output:** Weights $\theta_2, \dots, \theta_{T+1}$.

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1: for  $t = 1, \dots, T$  do
2:    $S_t \leftarrow \{i \in [n] : c_{t,i} = 1\}$  with independent  $c_{t,i} \sim \text{Bernoulli}(q)$ 
3:   for each  $i \in S_t$  with  $b_i = 1$  do
4:     sample  $y_{i,1}, \dots, y_{i,G} \sim \pi_{\theta_t}^\tau(\cdot | x_i)$  independently
5:      $r_{i,g} \leftarrow r(x_i, y_{i,g})$ ;  $A_{i,g} \leftarrow (r_{i,g} - \mu_i)/(s_i + c)$  as in (2)
6:      $g_i \leftarrow \nabla_{\theta} \ell_i(\theta_t)$  with  $\ell_i$  from (7)
7:      $\bar{g}_i \leftarrow g_i \min(1, C/\|g_i\|_2)$ 
8:      $z_t \sim \mathcal{N}(0, \sigma^2 C^2 I)$ ;  $\hat{g}_t \leftarrow (\sum_{i \in S_t} b_i \bar{g}_i + z_t)/(qn)$ 
9:    $\theta_{t+1} \leftarrow \text{Opt}(\theta_t, \hat{g}_t)$ 

```

would need a private tuning analysis [55]. Two execution effects lie outside the analysis: batched inference kernels and the training verifier’s wall-clock limit can couple one prompt’s rollouts and rewards to other records, and noise is generated in floating point rather than over the reals [56], [57]. Neither is quantified; Appendix A-G records the status of each assumption. (iii) *The bound is certified, not measured.* Every ϵ is a privacy-loss-distribution budget with pessimistic discretization, reproduced by an independent accountant and by Rényi accounting (Section VI-C, Appendix A-F), so the guarantee is as strong as the proof and the accounting make it. How much slack that bound leaves in practice, which an attack such as [3] run against the private model would measure, is left to future work (Section IX). (iv) *The curator is trusted.* A compromised curator, or a released copy of the recovery files, is outside the model. (v) *Not every private-looking quantity is a protected output.* The verifier is public code; rewards, completions, completion lengths, per-record gradients and system telemetry are training-time quantities that are never released and are not covered as outputs; and the bound concerns what the weights reveal about membership, not what the base model already knew about a prompt. The corpora of Section VII are public benchmarks used as a mechanism testbed; deployment on data whose subjects can be harmed requires a privacy unit aligned with those subjects.

V. THE DP-GRPO ALGORITHM

DP-GRPO is DP-SGD with the prompt group as the clipped record and the single-update GRPO loss as the per-record objective. Algorithm 1 defines it, and Figure 1 shows one round.

Line 6 uses the implemented single-update loss. With $u_{i,g,k} = \log \pi_{\theta}(y_{i,g,k} | x_i, y_{i,g,<k})$, completion length $L_{i,g}$, and the per-token KL estimate $K_{i,g,k}$ of (3),

$$\ell_i(\theta) = -\frac{1}{G} \sum_{g=1}^G \frac{1}{L_{i,g}} \sum_{k=1}^{L_{i,g}} \left[e^{u_{i,g,k} - \text{sg}(u_{i,g,k})} A_{i,g} - \beta K_{i,g,k} \right], \quad (7)$$

where sg stops gradients; advantages and reference log-probabilities are detached. This is (3) with one update per rollout batch, so the ratio equals one and the ratio clip is inactive; Appendix C records the implementation. Line 4 samples from π_{θ_t} at temperature $\tau = 0.7$; $\pi_{\theta_t}^\tau$ is a public function of θ_t . In line 5, s_i uses denominator $G - 1$ and $c = 10^{-4}$; constant-reward groups are retained with zero advantages. Rollouts are backpropagated in chunks that accumulate into g_i before line 7. Only LoRA coordinates are trainable. Line 8 runs even when S_t is empty, so qn is a public constant, and division by it, like line 9, is post-processing of the noisy sum.

The zero-advantage control sets $A_{i,g} = 0$ in (7), leaving the KL term and lines 7–9 intact. Its prompt contribution has the same norm bound, so both arms use the same accountant parameters. This intervention tests whether verifier-derived advantages contribute to utility after privatization.

VI. PRIVACY ANALYSIS

We now prove that Algorithm 1 satisfies Definition 3, the guarantee the threat model of Section IV requires.

A. Assumptions

The guarantee holds under four assumptions about how Algorithm 1 is run. The end of Appendix A records how each is enforced, tested, or left as a disclosed gap.

- (A1) *Public design.* Every public input of Algorithm 1, including all hyperparameters, is chosen without access to D .
- (A2) *Record isolation.* Given θ_t , the clipped gradients $\bar{g}_i$ of line 7 are independent across prompts, and the law of $\bar{g}_i$ depends only on (θ_t, x_i) .
- (A3) *Fresh hidden randomness.* The coins of line 2, the rollout randomness of line 4, and the noise of line 8 are drawn afresh in every round, mutually independent, independent of D , and never released.
- (A4) *Exact arithmetic.* Lines 4–9 are computed over the reals, and z_t is an exact Gaussian draw.

B. Guarantee

Write $A = A_{q,\sigma}$ and $B = B_\sigma$ for the pair of Definition 2.

Lemma 1 (One round of DP-GRPO): Assume (A1)–(A4). Fix a round t , any values of the prior releases $\hat{g}_{1:t-1}$, and prompt datasets $D \simeq D'$ such that slot k is present in D and absent in D' . Let M and M' be the conditional laws of $\hat{g}_t$ under D and D' . Then, for every $\alpha \geq 0$,

$$H_\alpha(M \| M') \leq H_\alpha(A \| B), \quad H_\alpha(M' \| M) \leq H_\alpha(B \| A). \quad (8)$$

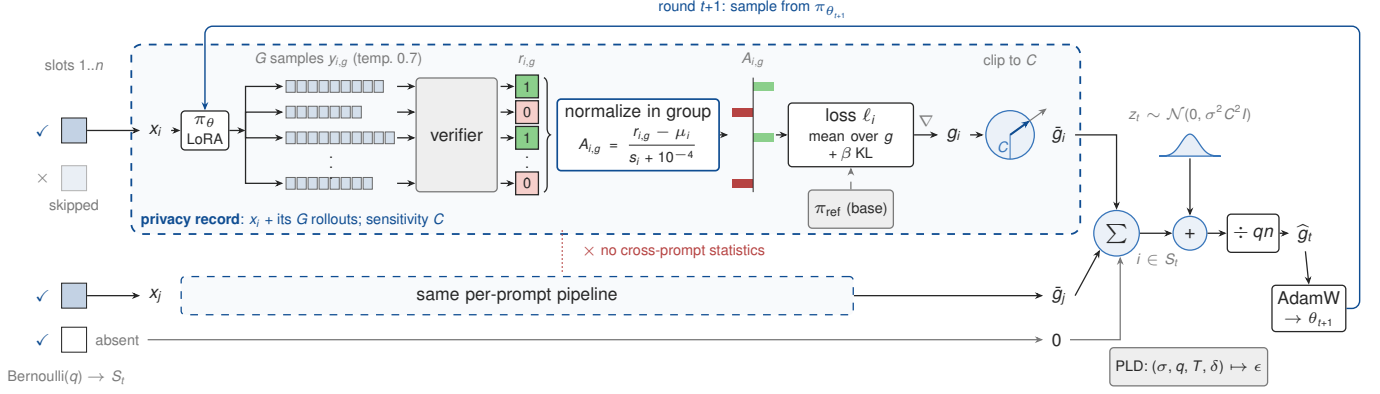


Fig. 1. One round of DP-GRPO (Algorithm 1). Each Poisson-sampled prompt x_i (line 2) generates G completions whose verifier rewards are normalized within that group only (lines 4–5). The rollout losses are combined into one prompt gradient g_i and clipped to norm C (lines 6–7), so the dashed region is a single privacy record. Clipped prompt gradients are summed over the sample, perturbed with Gaussian noise, divided by the public constant qn , and applied to the LoRA parameters (lines 8–9). The next round samples rollouts from the updated policy.

Conditioned on everything released so far, one round thus reveals no more about the prompt in slot k than one DP-SGD step does about a worst-case record, whatever t , the past releases, G , or C ; in the terminology of Zhu et al. [49], (A, B) dominates the round.

Theorem 1 (DP-GRPO is prompt-level DP): Assume (A1)–(A4). For every $\epsilon \geq 0$, Algorithm 1 is prompt-level $(\epsilon, \delta_{q,\sigma,T}(\epsilon))$ -DP (Definition 3).

Proof: Definition 3 requires (1) on every ordered pair of neighbors. Let $D \simeq D'$ with slot k present in D and absent in D' ; the reversed pair is handled by the same argument with the roles of A and B exchanged. By (A1), θ_t is a fixed public function of $\hat{g}_{1:t-1}$ (line 9 iterated from θ_1), so round t , conditioned on the prior releases, is the mechanism of Lemma 1, and (8) holds for every value of $\hat{g}_{1:t-1}$. The adaptive composition theorem for dominating pairs [49] (Lemma 4 in Appendix A) combines these per-round bounds into

$$H_{e^\epsilon}(\mathcal{L}_D(\hat{g}_{1:T}) \parallel \mathcal{L}_{D'}(\hat{g}_{1:T})) \leq H_{e^\epsilon}(A^{\otimes T} \parallel B^{\otimes T}) \leq \delta_{q,\sigma,T}(\epsilon),$$

and likewise with D, D' and A, B exchanged. By the characterization of (1) through H_{e^ϵ} (Section III-C), the releases $\hat{g}_{1:T}$ therefore satisfy (1) with $\delta = \delta_{q,\sigma,T}(\epsilon)$ on every neighboring pair. The trajectory $\theta_{2:T+1}$, the output named in Definition 3, is computed from $\hat{g}_{1:T}$ by the public maps of line 9, which do not read D , so Proposition 2 carries the bound to it; with T public by (A1), this is exactly the statement of Definition 3 with $\delta = \delta_{q,\sigma,T}(\epsilon)$. ■

Lemma 1 is proved in Appendix A, together with the two corollaries cited below. Any checkpoint selection that reads only the released weights and public data is further post-processing and inherits the guarantee. Three consequences of Theorem 1 organize the rest of the paper. First, $\delta_{q,\sigma,T}$ involves neither the group size G , the clipping norm C , the loss, nor the verifier, so G is a compute and utility parameter rather than a privacy cost. Second, because the proof reads lines 4–6 only through (A2)–(A4), any training procedure that keeps lines 2–3 and 7–9 and replaces lines 4–6 by a per-record computation inherits

the same guarantee at the same (q, σ, T) ; the verifier-filtered DP-SFT and gold-SFT comparators of Section VIII are such procedures (Corollary 2), so equal ϵ across arms means an identical mechanism, with the caveat that gold-SFT’s record carries the full reference solution rather than a final answer. Third, Definition 3 uses add/remove neighbors; replacing one prompt costs $(2\epsilon, (1 + e^\epsilon)\delta)$ by group privacy (Corollary 1).

C. Accounting

Every privacy budget in this paper is the profile $\delta_{q,\sigma,T}$ of Definition 2, evaluated by privacy-loss-distribution (PLD) accounting [26], [27] with the implementation in Google’s differential-privacy library [58] and the add-or-remove neighboring relation. The accountant discretizes the privacy loss by the connect-the-dots construction [28] with pessimistic rounding that assigns truncated tail mass to infinite loss, so every reported ϵ upper-bounds the budget of the ideal mechanism. Given a slot universe of size n , we set $\delta = 1/(10n)$ and $q = m/n$ for a chosen expected sample size m , fix the round count T , and take σ to be the smallest noise multiplier, found by bisection, at which the accountant returns the target ϵ . Two tasks with the same (n, m, T) and target therefore receive the same σ and the same certified budget, whatever their prompts, verifier, or loss. For every configuration used in this paper the accountant returns its target ϵ to at least four decimal places; Table VI in Appendix A lists each configuration with its certified budget and, for readers who reproduce our numbers with an RDP accountant, the corresponding Rényi-DP value. The implementation reproduces the published JAX-Privacy [59] reference value to within 3.1×10^{-4} (Appendix B).

VII. EXPERIMENTAL DESIGN

The experiments ask whether verifier feedback improves a policy trained under prompt-level differential privacy, whether private imitation at the same budget matches it, and how much of that comparison depends on answer scoring and on training choices. Within a benchmark, every compared arm spends the

same per-run privacy budget on the same prompts. A *design study* (Sec. VII-B) on MATH fixed the operating point and the supervised recipe; a *fixed protocol* (Sec. VII-C) then ran five arms on MATH and GSM8K, and on CommonsenseQA as an exploratory extension. MATH is therefore the development benchmark, whose fixed-protocol results replicate the design study, and GSM8K is the confirmatory one. Table IX (Appendix D) lists every arm, its seeds, and where it is reported, and Table VIII (Appendix C-F) lists every protocol decision and the runs it preceded or followed.

A. Common Setup

a) *Benchmarks.*: A benchmark supplies n training prompts, a deterministic verifier $r(x, y) \in \{0, 1\}$ that checks a response’s final answer against the prompt’s reference answer, and a disjoint evaluation set scored by the same verifier. The prompt with its reference answer is the private record (Sec. III-D); the verifier is public. MATH [60] and GSM8K [61] are the main benchmarks. CommonsenseQA [62], five-way multiple choice, tests the comparison outside mathematics; its 6,961 training questions are sampled so that n , and with it every mechanism parameter, equals GSM8K’s, and it is evaluated on the 1,221-question validation split because test labels are not public. Its verifier reads one option letter from an “Answer: {letter}” declaration, a format adopted after a pilot with a stricter answer marker failed. SVAMP [63] is used only to evaluate GSM8K checkpoints.

b) *Model and training.*: Every arm fine-tunes rank-4 LoRA adapters on Qwen2.5-1.5B-Instruct (configuration in Appendix C). GRPO draws $G=8$ completions per prompt at temperature 0.7 with KL weight $\beta=0.04$. The learning rate is 3×10^{-4} for DP-GRPO and the zero-advantage control, chosen by a pilot on MATH and carried unchanged to the other benchmarks; the supervised arms use the learning rates and clipping norm of their own recipes, selected as described in Appendix C-D. Each run is one (ϵ, δ) -DP release of the LoRA weights, with σ set by bisection so that T rounds compose to the target ϵ (Sec. VI-C); Table VI certifies the budget of every configuration. The budget is per run; pilot tuning and any released randomness are outside it (Sec. IV).

c) *Shared and per-benchmark settings.*: Every benchmark uses $\epsilon=8$ at $\delta=1/(10n)$, $qn=128$ expected prompts per round, $G=8$, and $C=0.055$ for DP-GRPO and its control (the supervised arms’ norm is in Appendix C-B). Within a benchmark all arms share T , q , σ and the training prompts, and may differ in C without changing ϵ because the noise scales with C (Corollary 2); for the same reason, arms with different C also share the ratio of noise to the largest clipped signal, and C sets only how much of each record’s gradient survives clipping. Table I lists what differs: the horizon ($T=50$ on MATH, $T=100$ on the other two), the σ it implies, and a completion cap matched to solution length; these are the parameters every result in Section VIII refers to. The horizons are short by design: at a fixed ϵ more rounds require a larger σ , and 50 to 100 rounds keep σ near 0.5 while spending the budget on under two expected passes over the prompts (qT in Table I).

TABLE I
PER-BENCHMARK SETTINGS OF THE PRIVATE RUNS. SHARED BY ALL THREE: $\epsilon=8$ AT $\delta=1/(10n)$, $qn=128$, $G=8$, $C=0.055$ (DP-GRPO, CONTROL), LEARNING RATE 3×10^{-4} . FULL-PRECISION VALUES, THE SUPERVISED ARMS’ RECIPE PARAMETERS AND SPLIT CONSTRUCTION ARE IN APPENDIX C; CERTIFIED BUDGETS IN TABLE VI.

	MATH	GSM8K	CSQA
Role	development	confirmatory	exploratory
Training prompts n	6,988	6,961	6,961
Evaluation set	4,499 test	1,319 test	1,221 val.
Rounds T	50	100	100
Expected passes qT	0.92	1.84	1.84
Noise multiplier σ	0.484	0.520	0.520
Completion cap (tokens)	1,024	512	512
Strict scorer	boxed answer	final number	option letter

GSM8K’s protocol, including its horizon, was fixed before any private GSM8K run. We set $\delta=1/(10n)$, an order of magnitude below $1/n$.

B. Stage 1: Design Study

MATH was the development benchmark, and every design run used two initialization seeds per setting. The *budget sweep* sets $\epsilon \in \{1, 4.23, 8\}$ at fixed T by changing σ alone; these runs come from the earlier design study and measure the utility of the procedure at each noise level (Appendix C-A). The *clipping sweep* sets $C \in \{\frac{1}{2}, 1, 2\} \times 0.055$ at fixed ϵ , which moves clipping bias and absolute noise together. The *allocation study* replaces eight rollouts on 128 expected prompts per round with two rollouts on 512 (the same rollout count) or on 128 (a quarter of it); each change also changes q and σ . Two *supervised challenges* run filtered DP-SFT at DP-GRPO’s learning rate (same-LR) and at $T=100$ with σ recalibrated to keep $\epsilon=8$ (100-update). Finally, *non-private references* remove the noise ($\sigma=0$) at the private hyperparameters, with clipping kept on MATH, kept or disabled on GSM8K, and disabled on CommonsenseQA; they make no privacy claim, and the GSM8K references, as the one exception to final-step evaluation, are scored at steps 25, 50 and 100. A two-seed clipped MATH reference and a single-seed unclipped GSM8K reference, both at learning rate 10^{-4} , test a smaller step. Appendix D-B reports the budget, clipping and non-private results and Section VIII-B the supervised challenges.

C. Stage 2: Fixed Protocol

Five arms run on each benchmark at that benchmark’s settings. The *public base* is the untuned model. *DP-GRPO* is Algorithm 1. The *zero-advantage control* runs the identical private pipeline (generation, verification, KL term, clipping, noise and optimizer updates) but sets every advantage in Eq. (2) to zero before the loss, so correct responses receive no preference over incorrect ones. It is a trained control, not the public base; it shares DP-GRPO’s procedure and hyperparameters but not its realized noise or rollouts, and the difference between the two measures the reward-advantage term. The LoRA update starts at zero (PEFT’s default initialization of B), so the control starts at the reference policy, where the

KL gradient vanishes: its first update is pure noise, and later updates add only the KL pull toward the base. The control therefore answers the question the private setting raises first: whether the private pipeline itself, its noise, clipping and KL pull, moves the model without a reward signal, so that a gain of DP-GRPO over the base can be attributed to the reward-advantage term rather than to private training as such. A control close to the base is the expected outcome, and the contrast with DP-GRPO is then the reward effect as a whole; which part of the advantage carries it is the finer question that the advantage ladder of Section IX would take up.

Two supervised arms train the same LoRA with prompt-level DP-SGD at the same q , σ and T , hence the same ϵ , following [11], [24], without a KL term. *Verifier-filtered DP-SFT* adapts rejection-sampling fine-tuning [38]: it samples $G=8$ public-base completions per prompt, keeps those the verifier accepts, and minimizes their negative log-likelihood averaged within the prompt before clipping; a prompt with no accepted completion contributes a zero gradient. *Gold DP-SFT* trains on the reference target, which DP-GRPO never sees: a worked solution on MATH and GSM8K, and the four-token declaration “Answer: {gold letter}” on CommonsenseQA.

a) Seeds.: Every arm has two initialization seeds per benchmark. On MATH and GSM8K, DP-GRPO and the control have two more, added after the two-seed results had been inspected, so their four-seed contrasts are sequential, exploratory extensions. Supervised contrasts pair the two SFT seeds with DP-GRPO seeds 1–2, which share their initializations. Per-seed values are in Appendix D-A.

D. Endpoints and Scoring

The *primary endpoint* is strict accuracy of one greedy completion per evaluation problem at the final training step, with no checkpoint or problem selection and a scorer frozen before the arms were run. Strict scoring credits a correct extracted answer only: a boxed answer checked by symbolic equality on MATH, a final number on GSM8K, an option letter on CommonsenseQA. A response with no extraction counts as incorrect, and every problem stays in the denominator. MATH is scored on 4,499 test problems, excluding those that earlier runs used for checkpoint selection (Appendix C); GSM8K on its full test split.

Sensitivity analyses rescore the same saved responses and change no training reward. On MATH, the *format-robust* endpoint also credits a completed unboxed final answer under the same symbolic check; responses cut at the cap stay ineligible, and the rules were audited before arm-level rescoring (Appendix E). On CommonsenseQA, a *final-answer* rescore reads the last explicit answer declaration and abstains on ambiguous ones. It was chosen after the strict scores of four arms had been inspected, then hash-frozen before their rescoring and before filtered-SFT evaluation, so it is a post-hoc sensitivity endpoint (Appendix E-B). GSM8K’s scorer already credits an unboxed final number, so it has no separate format-robust endpoint. For transfer, the final GSM8K checkpoints are scored on SVAMP’s 1,000 problems with GSM8K’s prompt and

cap (Appendix D-D). Every compared arm, the base included, is scored in the same evaluation session, and accuracies are never compared across benchmarks.

E. Statistics

A contrast is the paired difference in accuracy between two arms on the shared evaluation set, in percentage points (pp). Brackets give a 95% problem-cluster bootstrap interval over the evaluation set (100,000 draws; each arm’s seeds averaged within a problem); the four-seed contrasts additionally report the standard deviation of the per-seed differences and a seed-level t interval. Each contrast also carries exact McNemar tests on each seed pair, Holm-adjusted within its family, and a sign-randomization p -value for the pooled difference. A family is the set of seed-pair tests of one contrast on MATH and GSM8K, and the eight tests of the final-answer rescore on CommonsenseQA, where the rescore was fixed after the arms had been seen. Beside every contrast whose interval includes zero we give the two-sided 80%-power minimum detectable effect (MDE), so such a result reads as a bound on the effect rather than as its absence.

VIII. RESULTS

MATH and GSM8K provide the main evidence; CommonsenseQA tests a non-mathematical setting, and SVAMP tests transfer without further training. Fig. 2 places every arm against the public base, and Table II gives DP-GRPO’s gain over each comparator (positive values favor DP-GRPO). Strict scoring is primary throughout; arms, scorers, intervals and MDEs are as defined in Section VII, every private arm runs at the parameters of Table I, and its certified budget is in Table VI.

A. Reward Learning Survives Prompt-Level Privacy

At the per-run $\epsilon=8$ operating point, DP-GRPO improves strict accuracy over the public base by +2.51 [1.54, 3.48] pp on MATH and +3.60 [1.76, 5.44] on GSM8K, reaching four-seed means of 51.6% and 74.3% from bases of 49.1% and 70.7%. The zero-advantage control, which trains through the same private pipeline without any reward preference, stays at the base, -0.14 [-0.65 , 0.37] and $+0.36$ [-0.53 , 1.27] with MDEs 0.73 and 1.29, as expected for a pipeline whose only remaining signal is the KL pull toward the base (Section VII-C): noise and clipping alone neither help nor harm at this budget. DP-GRPO exceeds it by +2.65 [1.79, 3.51] on MATH and +3.24 [1.61, 4.89] on GSM8K, so within this pipeline the gain is attributable to the reward-advantage term rather than to private training as such.

The effect is consistent across training runs. Every seed-level reward contrast is positive on both benchmarks, and every seed-level McNemar test rejects equality after Holm adjustment, the largest adjusted p being 0.013 on GSM8K; the sign-randomization p for the pooled contrast is below 10^{-3} on both. The per-seed differences have a standard deviation of 0.36 pp on MATH and 0.72 on GSM8K, and seed-level t intervals of [2.08, 3.22] and [2.10, 4.38], so the reward effect is seven and four times its seed-to-seed spread. These pools

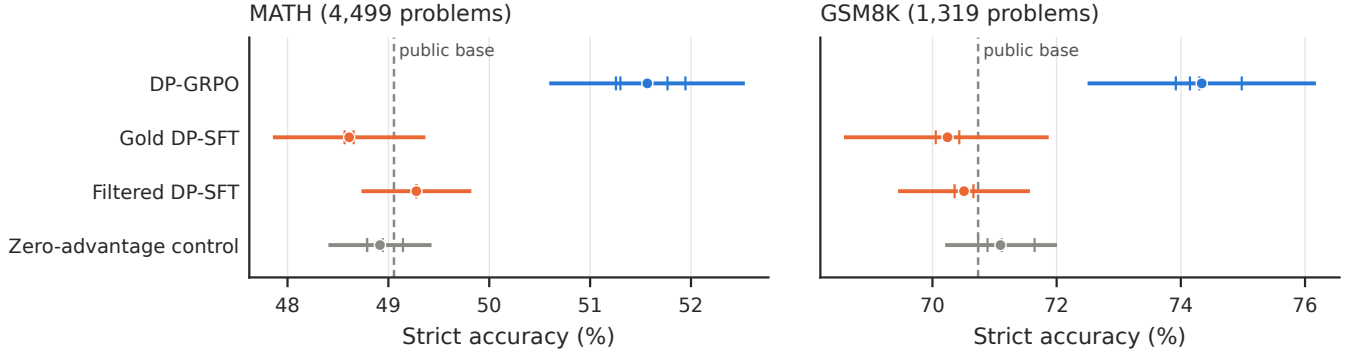


Fig. 2. Strict accuracy of each arm at $\epsilon=8$. Dot: mean over seeds; ticks: individual seeds; dashed line: public base. Each bar is the 95% paired interval of the arm’s difference from the base, added to the base value, so a bar that clears the dashed line is a resolved gain and one that crosses it is not. Exact values in Appendix D-A.

TABLE II
DP-GRPO ACCURACY GAIN OVER EACH COMPARATOR.

Comparator	MATH (pp)	GSM8K (pp)
Zero-adv. control	+2.65 [1.79, 3.51]	+3.24 [1.61, 4.89]
Filtered DP-SFT	+2.32 [1.39, 3.26]	+3.53 [1.74, 5.31]
Gold DP-SFT	+2.99 [2.01, 3.98]	+3.79 [1.97, 5.57]

Values are strict-accuracy differences in percentage points (pp), with 95% paired CIs. Positive values favor DP-GRPO. The control row pools four DP-GRPO and four control seeds, two of each added sequentially (Sec. VII-C); the supervised rows pair two seeds each.

include the two sequentially added seeds (Sec. VII-C); per-seed contrasts and tests are in Appendix D-A.

B. Comparison with Private Supervised Fine-Tuning

DP-GRPO outperforms both private imitation procedures on MATH and GSM8K (Table II). All eight seed-paired supervised contrasts, two seed pairs for each of the four task-by-target comparisons, favor it and are significant after Holm adjustment, with pooled gains from 2.32 to 3.79 pp. The supervised arms themselves stay within their MDEs of the base. Filtered DP-SFT changes MATH by +0.22 [−0.32, 0.77] and GSM8K by −0.23 [−1.29, 0.83] (MDEs 0.78 and 1.50); gold DP-SFT, despite training on the worked reference solutions as token-level targets, which DP-GRPO touches only through the verifier’s answer check, changes them by −0.44 [−1.20, 0.31] and −0.49 [−2.16, 1.14] (MDEs 1.08 and 2.36). The same supervised pipeline does learn when its targets are short: on CommonsenseQA, gold DP-SFT gains +4.95 pp over the base and matches DP-GRPO (Section VIII-D). The flat mathematical results therefore reflect what these recipes extract from long worked solutions under this noise, not a pipeline that fails to train; Section IX discusses why. These results compare specified procedures at a matched per-run budget and guarantee (recipe selection in Appendix C-D); they concern these recipes rather than private imitation in general.

Two MATH challenges test whether the gap closes at DP-GRPO’s learning rate or with twice as many updates at recalibrated noise. Neither does: DP-GRPO remains +2.10

TABLE III
MATH ACCURACY UNDER TWO ANSWER EXTRACTORS.

Model	Strict (%)	Format-robust (%)
Public base	49.06	50.37
Zero-adv. control	48.92 (−0.14)	50.28 (−0.08)
Filtered DP-SFT	49.28 (+0.22)	50.47 (+0.10)
Gold DP-SFT	48.61 (−0.44)	49.98 (−0.39)
DP-GRPO	51.57 (+2.51)	51.62 (+1.26)
Non-private GRPO [†]	52.12 (+2.96)	—

Parentheses give the difference from the public base in percentage points on 4,499 problems. Both extractors score the same saved responses. [†]Earlier run of standard GRPO, one seed: LoRA rank 16, LR 5×10^{-6} , checkpoint 900, with its gain measured against its own base (49.17%; Appendix C-A); later checkpoints score 51.4–52.7%. It was not format-robustly rescored.

TABLE IV
GSM8K ACCURACY.

Model	Accuracy (%)
Public base	70.74
Zero-adv. control	71.10 (+0.36)
Filtered DP-SFT	70.51 (−0.23)
Gold DP-SFT	70.24 (−0.49)
DP-GRPO	74.34 (+3.60)
Non-private GRPO [†]	75.36 (+4.02)

Parentheses give the difference from the public base in percentage points on 1,319 questions. The private arms use 100 updates; DP-GRPO and the zero-advantage control average four seeds, the supervised arms two. [†]Earlier run of standard GRPO, one seed: LoRA rank 16, LR 5×10^{-6} , two epochs (1,740 updates). Its gain is measured against its own base (71.34%; Appendix C-A).

[1.14, 3.07] above the same-LR arm and +1.78 [0.88, 2.67] above the 100-update arm. The longer run is the only supervised change with a resolved improvement, +0.54 [0.03, 1.06] over the selected recipe (Table XII).

C. Answer Formatting and Correctness

Two robustness checks separate answer format and response length from correctness. Tables III and IV give the MATH and

TABLE V
COMMONSENSEQA ACCURACY UNDER TWO ANSWER EXTRACTORS.

Model	Strict (%)	Final answer (%)
Public base	46.19	66.75
Zero-adv. control	45.99 (−0.20)	67.20 (+0.45)
Filtered DP-SFT	52.42 (+6.22)	66.71 (−0.04)
Gold DP-SFT	71.70 (+25.51)	71.70 (+4.95)
DP-GRPO	70.84 (+24.65)	71.46 (+4.71)
Non-private GRPO	77.27 (+31.08)	77.27 (+10.52)

Parentheses give the difference from the public base in percentage points. Both extractors score the same saved responses on 1,221 questions. The non-private reference keeps DP-GRPO’s hyperparameters and seeds and sets $\sigma = 0$ without clipping.

GSM8K accuracies; their non-private rows are discussed in Section VIII-E.

The reward effect survives the format-robust scorer. The reward contrast is +1.34 [0.54, 2.14] pp, positive in all four seeds, and the gain over the base +1.26 [0.33, 2.17]. Both are smaller than under strict scoring because the rescore lifts the base from 49.06% to 50.37% and the control from 48.92% to 50.28%, while DP-GRPO, whose correct answers are nearly all boxed already, moves only to 51.62%. Table XIV (Appendix E) locates the gain. On the 3,831 problems the base already answers in the required form, DP-GRPO changes accuracy by −0.24 pp; on the 668 problems the base leaves without a boxed answer it adds +1.49 pp under format-robust scoring and +2.75 under strict. The reward signal thus teaches the model to produce a checkable answer on problems where the base produces none, while leaving the problems the base already answers essentially unchanged. DP-GRPO also stays resolved above selected-LR filtered and gold DP-SFT, by +1.19 [0.30, 2.08] and +1.68 [0.74, 2.61]; against the 100-update challenge the gap is +0.78 [−0.07, 1.62], MDE 1.22, so that comparison depends on the scorer.

GSM8K needs no separate rescoring: its scorer already credits an unboxed final number, so its strict endpoint is format-robust by construction, and its reward contrast, +3.24 [1.61, 4.89] pp in every seed, is the format-independent evidence for C1. Nor does the MATH gain come from shorter responses: DP-GRPO’s responses average 554 tokens against the base’s 540.

D. Outside Mathematics: CommonsenseQA

On CommonsenseQA the strict gain over base, +24.7 [22.1, 27.3] pp, comes mostly from extraction: the strict extractor recognizes only 64% of base responses, partly because discussion of several options reads as conflicting answers. Under the final-answer rescore (Table V), a resolved reward effect remains. DP-GRPO reaches 71.5% against 66.7% for base and 67.2% for the control, with gaps of +4.71 [2.87, 6.59] over base, +4.26 [2.50, 6.06] over the control, and +4.75 [2.99, 6.55] over filtered DP-SFT, each with Holm-adjusted $p < 0.001$ within the eight-test family. Even crediting all 21 remaining base abstentions as correct leaves a descriptive 2.99-point gap below DP-GRPO; this bound does not cover errors

in accepted answers (Appendix E-B). Gold DP-SFT, whose targets are four-token answer declarations, reaches 71.7%, and the DP-GRPO–gold gap is −0.25 [−1.56, 1.02]. The reward effect thus carries beyond mathematics. That gold-label SFT matches it here is what the target structure predicts: a four-token declaration supervises exactly the scored answer, so imitation and reward optimization have the same thing to learn from each record, whereas a mathematical reference solution spends most of its tokens on the derivation, so under the same clipped and noised gradient the answer receives a smaller share of the signal (Section IX).

E. Comparison with Non-Private Training

Each benchmark table ends with one non-private GRPO run. On MATH and GSM8K it is standard GRPO without privacy, scored against its own base (configuration in Appendix C-A); on CommonsenseQA it keeps DP-GRPO’s hyperparameters and seeds and removes the noise and the clipping. Measured against each run’s own base, DP-GRPO retains most of the non-private gain on the mathematical tasks: +2.51 against +2.96 pp on MATH (85%) and +3.60 against +4.02 on GSM8K (90%). On CommonsenseQA it retains 45%, +4.71 against +10.52 under the final-answer rescore. Only this last comparison is paired: the non-private run exceeds DP-GRPO by +6.43 [4.67, 8.23] pp strict and +5.81 [4.10, 7.58] under the rescore, a resolved privacy cost, though measured at the private hyperparameters rather than against a tuned ceiling. The mathematical-task ratios are descriptive and carry no interval; only the CommonsenseQA comparison is paired.

F. Supporting Studies

On MATH, accuracy is insensitive to the budget over the tested range: moving from $\epsilon=8$ to $\epsilon=1$ changes it by −0.57 [−1.19, 0.06] pp, so an eightfold tighter guarantee costs at most 1.2 points, and halving or doubling the clipping norm changes it by less than its MDE (Appendix D-B). The $\epsilon=1$ run doubles the noise multiplier and still reaches the same accuracy, which fits the account in Section IX: the attainable gain on MATH is small, and DP-GRPO reaches most of it under either noise level. Drawing two rollouts per prompt instead of eight cuts measured training time $3.2\times$ at a strict-accuracy cost of 0.96 [0.18, 1.74] pp, about half of which is answer formatting, since the cost falls to 0.49 [−0.27, 1.26] under format-robust scoring (Appendix D-C). Transfer of the GSM8K checkpoints to SVAMP without further training gains +1.20 [−0.53, 2.95] pp over base, positive in all four seeds; with 1,000 problems the MDE is 2.49 pp, so a transfer gain of GSM8K’s size would have been detected and one of a third of that size, which the point estimate shows, would not (Appendix D-D).

IX. DISCUSSION AND LIMITATIONS

a) *When imitation is effective.*: The supervised comparisons fit one account. Filtered imitation can only reinforce prompts the public base already answers correctly, whereas reward optimization exploits contrasts among responses from the evolving policy, including prompts the base answers

inconsistently. Gold imitation supplies a target for every prompt, but what that target supervises decides how much of the clipped, noised gradient reaches the scored answer: CommonsenseQA’s four-token declarations supervise the answer and nothing else, so gold DP-SFT matches DP-GRPO there, while a mathematical reference solution spends most of its tokens on the derivation, so the answer receives a small share of a gradient whose norm is capped at C . A target-length study at fixed answer supervision would measure this share directly.

b) When privacy costs accuracy.: The cost of privacy appears to track how much a task leaves to learn (Section VIII-E). On MATH and GSM8K the non-private runs gain only three to four points over the base, and under its recipe the MATH run stays between 51.4% and 52.7% from 900 to 3,494 updates, so training longer without privacy does not raise that target. DP-GRPO reaches most of this small gain within 50 or 100 noisy updates. On CommonsenseQA the non-private run gains 10.5 points under the final-answer rescore, and DP-GRPO captures less than half of it in 100 updates. A plausible mechanism is the noise. On MATH and GSM8K its norm is eight to nine times the largest possible clipped signal (Appendix C-A), and under AdamW it shortens each step along the signal. Shorter steps cost little when the attainable gain is small, as on the mathematical tasks, and more when there is more to learn, as on CommonsenseQA, whose answers of about 100 tokens never reach the cap. Our experiments do not isolate this explanation.

The small non-private gains also mark a limit of our setting. Every run uses one 1.5B-parameter model, and its strength likely bounds what reward learning of any kind, private or not, can add on these benchmarks. A stronger base model may have more to gain, and whether privacy then costs more, as on CommonsenseQA, or stays small is untested. Within this bound the method is useful: under a prompt-level guarantee, DP-GRPO keeps most of the non-private gain on both mathematical tasks, varies little across seeds, and improves on the matched control on all three tasks.

c) Scope and inference.: All results come from one model, Qwen2.5-1.5B-Instruct with rank-4 LoRA, whose strength may bound the attainable gains (see above). The evidence for C1 and C2 comes from two mathematical-reasoning benchmarks whose verifiers check a final answer; CommonsenseQA serves only to check that the reward effect carries beyond mathematics (Sec. VIII-D). The design study ran on MATH, with only non-private references on GSM8K and CommonsenseQA, so the privacy-budget, clipping and allocation results describe one benchmark. Transfer was checked once, by scoring the GSM8K checkpoints without further training on SVAMP [63] (Appendix D-D): DP-GRPO improves on the base there by a third of its GSM8K gain, within the interval over problems, and the advantage over DP-SFT is not detectable, so C2 is an in-distribution claim. The budget and clipping-norm nulls exclude large effects over the tested ranges, not small ones (Appendix D-B).

d) Measurement.: The verifier defines correctness, and strict scoring also rewards answer format. Under the format-robust endpoint the MATH DP-GRPO–base gain falls from

2.51 to 1.26 points (Table III); GSM8K’s scorer already credits unboxed answers, so it has no such endpoint. The MATH evaluation set excludes the 500 problems that earlier runs used for checkpoint selection (Appendix C), and the earlier design-study runs in Table XII are compared only with each other (Appendix C-A).

e) Comparators.: The supervised arms use recipe parameters selected on MATH and carried unchanged to the other benchmarks (Appendix C-D), so C2 is a statement about these recipes at a matched guarantee. The non-private references keep the private hyperparameters rather than being tuned for noiseless training, so the CommonsenseQA cost is measured against a reference that a tuned run could raise, and the mathematical-task ratios are descriptive (Section VIII-E); a non-private learning-rate sweep would turn both into interval estimates. The horizon differs between benchmarks ($T=50$ and 100).

f) Future work.: Each of these gaps has a direct experiment. A one-run audit with enough canaries to have power at this horizon [29] would put an empirical lower bound beside the proof, and a membership attack [3] run on both the non-private and the private model would show how much measurable leakage the mechanism removes. The zero-advantage control extends naturally into an advantage ladder: adding a shuffled-reward arm, which keeps the advantage distribution but detaches it from correctness, and a positive-only unnormalized advantage that scores each verified completion 1, between the control and the full group-normalized advantage, would show how much of the reward effect comes from verifier correctness and how much from the within-group contrast itself, refining the attribution the control already provides. A compute-matched DP-SFT with its pilot budget charged to ϵ [55] would make the supervised comparison end to end. Verifiers beyond final-answer mathematics, such as unit tests for code, would test whether the result depends on short binary rewards. Finally, moving the record from one prompt to one contributor’s prompts by clipping at the user boundary [8] is the step that a corpus whose subjects can be harmed would require (Sec. IV).

X. CONCLUSION

Prompt-level privacy for RLVR depends on choosing the record: taking the group of responses to one prompt as the record makes GRPO an instance of DP-SGD whose budget depends on neither the group size nor the clipping norm, and yields, to our knowledge, the first differential privacy guarantee for RLVR training. Under that guarantee, at a per-run budget of $\epsilon = 8$ on one 1.5B-parameter model, the verifier’s reward signal still improves accuracy on MATH and GSM8K over a matched control in every seed, the improvement survives format-robust rescoring and length controls, and at the same budget it outperforms the evaluated private imitation recipes on both tasks. DP-GRPO retains 85–90% of the non-private gain on the mathematical tasks, where the noise of an eightfold tighter budget changes MATH accuracy by at most 1.2 points, and less than half on CommonsenseQA, where more remains

to learn. The protocol that separates these effects, a zero-advantage control at a matched budget with format-robust and response-length checks, applies to any private post-training method. Larger base models, which leave more to learn, are the natural next test of both the method and its cost.

LLM USAGE CONSIDERATIONS

LLMs were used for editorial purposes in this manuscript, and all outputs were inspected by the authors to ensure accuracy and originality. We also used an LLM (Anthropic’s Claude) beyond editing: it drafted and revised prose, helped write and review the training, evaluation and analysis code, and produced the tables and figures from the saved outputs. The authors set the research questions, the mechanism, the experimental design and every claim in the paper; reviewed the code, whose privacy-critical paths are covered by the mechanism-verification tests of Appendix B; approved every job before it ran; and checked all reported numbers and references against their sources. The model we train, Qwen2.5-1.5B-Instruct, is the object of study and contributes no text or judgment to this paper. Every run trains on a single GPU, the longest for about 12 hours (Appendix C-A). The authors take full responsibility for the paper.

ETHICAL CONSIDERATIONS

All training and evaluation data are public benchmarks, and no human subjects were involved. We treat the training prompts as private only to evaluate the mechanism. The guarantee protects one prompt, not one person: a person who contributes k prompts is protected only through group privacy, which degrades with k (Proposition 1). A deployment on confidential problems whose authors or subjects could be harmed therefore needs a privacy unit aligned with them, such as user-level accounting [8]. The mechanism releases weights only; the training telemetry it reads must stay unreleased.

REFERENCES

- [1] N. Lambert, J. Morrison, V. Pyatkin, S. Huang, H. Ivison, F. Brahman, L. J. V. Miranda, A. Liu, N. Dziri, X. Lyu, Y. Gu, S. Malik, V. Graf, J. D. Hwang, J. Yang, R. Le Bras, O. Tafjord, C. Wilhelm, L. Soldaini, N. A. Smith, Y. Wang, P. Dasigi, and H. Hajishirzi, “Tulu 3: Pushing frontiers in open language model post-training,” in *Conference on Language Modeling (COLM)*, 2025.
- [2] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. K. Li, Y. Wu, and D. Guo, “DeepSeekMath: Pushing the limits of mathematical reasoning in open language models,” *arXiv:2402.03300*, 2024.
- [3] Y. Liu, H. Zhang, J. Zheng, Z. Sun, Z. Peng, J. Wei, T. Cong, Y. Yang, and X. He, “Auditing data membership in reinforcement learning with verifiable rewards,” *arXiv:2511.14045*, 2026.
- [4] J. Zhang, M. Lei, M. Ding, M. Li, Z. Xiang, D. Xu, J. Xu, and D. Wang, “Towards user-level private reinforcement learning with human feedback,” in *Conference on Language Modeling (COLM)*, 2025, arXiv:2502.17515.
- [5] Y. H. Cho and W. W. Sun, “Privacy-preserving reinforcement learning from human feedback via decoupled reward modeling,” *Statistical Learning and Data Science*, 2026, Art. no. 100021.
- [6] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, “Deep learning with differential privacy,” in *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2016, pp. 308–318.
- [7] F. Wu, H. A. Inan, A. Backurs, V. Chandrasekaran, J. Kulkarni, and R. Sim, “Privately aligning language models with reinforcement learning,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [8] Z. Charles, A. Ganesh, R. McKenna, H. B. McMahan, N. Mitchell, K. Pillutla, and K. Rush, “Learning with user-level differential privacy under fixed compute budgets,” in *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2025, pp. 901–920.
- [9] L. Chua, B. Ghazi, Y. Huang, P. Kamath, R. Kumar, D. Liu, P. Manurangsi, A. Sinha, and C. Zhang, “Mind the privacy unit! User-level differential privacy for language model fine-tuning,” in *Conference on Language Modeling (COLM)*, 2024.
- [10] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *International Conference on Learning Representations (ICLR)*, 2022.
- [11] D. Yu, S. Naik, A. Backurs, S. Gopi, H. A. Inan, G. Kamath, J. Kulkarni, Y. T. Lee, A. Manoel, L. Wutschitz, S. Yekhanin, and H. Zhang, “Differentially private fine-tuning of language models,” in *International Conference on Learning Representations (ICLR)*, 2022.
- [12] K. Chen, H. Tang, Q. Liu, and Y. Xu, “Improved algorithms for differentially private language model alignment,” *arXiv:2505.08849*, 2025.
- [13] A. Rio, M. Barlier, and I. Colin, “Differentially private policy gradient,” *arXiv:2501.19080*, 2025.
- [14] D. Yu, P. Kairouz, S. Oh, and Z. Xu, “Privacy-preserving instructions for aligning large language models,” in *International Conference on Machine Learning (ICML)*, 2024, arXiv:2402.13659.
- [15] R. Zhang and N. Mureshghallah, “Reinforcement learning on benign facts amplifies leakage of memorized private data,” *arXiv:2608.21727*, 2026.
- [16] Y. Zhang, K. Panaganti, L. Shi, J. Ziani, and A. Wierman, “KL-regularization itself is differentially private in bandits and RLHF,” *arXiv:2505.18407*, 2025.
- [17] Y. He and X. Zhou, “On the sample complexity of differentially private policy optimization,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [18] —, “Towards differentially private reinforcement learning with general function approximation,” *arXiv:2605.07049*, 2026.
- [19] F. Khadem, S. Mousavi, Y. Fang, and Y. Liu, “DP-OPD: Differentially private on-policy distillation for language models,” *arXiv:2604.04461*, 2026.
- [20] A. Khadangi, A. Sartipi, I. Tchappi, R. Bahmani, and G. Fridgen, “Efficient differentially private fine-tuning of LLMs via reinforcement learning,” *arXiv:2507.22565*, 2025.
- [21] Y. Hu, R. McKenna, D. Yu, S. Wu, H. Zhao, Z. Xu, and P. Kairouz, “ACTG-ARL: Differentially private conditional text generation with rl-boosted control,” in *International Conference on Machine Learning (ICML)*, 2026.
- [22] S. Ghiasvand and N. Schatbakhsh, “GRASP: Reinforcing language model anonymizers with group relative policy optimization,” *arXiv:2608.06526*, 2026.
- [23] H. Zhou, K. Ye, E. Xu, J. Zhu, Y. Yang, S. Gong, and C. Shi, “Demystifying group relative policy optimization: Its policy gradient is a u-statistic,” *CoRR*, vol. abs/2603.01162, March 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2603.01162>
- [24] X. Li, F. Tramer, P. Liang, and T. Hashimoto, “Large language models can be strong differentially private learners,” in *International Conference on Learning Representations*, 2022. [Online]. Available: <https://openreview.net/forum?id=bVuP3ltATMz>
- [25] S. De, L. Berrada, J. Hayes, S. L. Smith, and B. Balle, “Unlocking high-accuracy differentially private image classification through scale,” *arXiv:2204.13650*, 2022.
- [26] D. M. Sommer, S. Meiser, and E. Mohammadi, “Privacy loss classes: The central limit theorem in differential privacy,” *Proceedings on Privacy Enhancing Technologies (PoPETs)*, vol. 2019, no. 2, pp. 245–269, 2019.
- [27] A. Koskela, J. Jälkö, and A. Honkela, “Computing tight differential privacy guarantees using FFT,” in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020.
- [28] V. Doroshenko, B. Ghazi, P. Kamath, R. Kumar, and P. Manurangsi, “Connect the dots: Tighter discrete approximations of privacy loss distributions,” *Proceedings on Privacy Enhancing Technologies (PoPETs)*, vol. 2022, 2022.
- [29] T. Steinke, M. Nasr, and M. Jagielski, “Privacy auditing with one (1) training run,” in *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. [Online]. Available: <https://openreview.net/forum?id=f38EY21IBw>
- [30] W. Zeng, Y. Huang, Q. Liu, W. Liu, K. He, Z. Ma, and J. He, “SimpleRL-Zoo: Investigating and taming zero reinforcement learning for open base

- models in the wild,” in *Conference on Language Modeling (COLM)*, 2025.
- [31] Y. Wang, Q. Yang, Z. Zeng, L. Ren, L. Liu, B. Peng, H. Cheng, X. He, K. Wang, J. Gao, W. Chen, S. Wang, S. S. Du, and Y. Shen, “Reinforcement learning for reasoning in large language models with one training example,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [32] Y. Yue, Z. Chen, R. Lu, A. Zhao, Z. Wang, Y. Yue, S. Song, and G. Huang, “Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model?” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. [Online]. Available: <https://openreview.net/forum?id=4OsgYD7em5>
- [33] R. Shao, S. S. Li, R. Xin, S. Geng, Y. Wang, S. Oh, S. S. Du, N. Lambert, S. Min, R. Krishna, Y. Tsvetkov, H. Hajishirzi, P. W. Koh, and L. Zettlemoyer, “Spurious rewards: Rethinking training signals in RLVR,” in *International Conference on Machine Learning (ICML)*, 2026.
- [34] M. Wu, Z. Zhang, Q. Dong, Z. Xi, J. Zhao, S. Jin, X. Fan, Y. Zhou, H. Lv, M. Zhang, Y. Fu, Q. Liu, S. Zhang, and Q. Zhang, “Reasoning or memorization? Unreliable results of reinforcement learning due to data contamination,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026.
- [35] A. Hochlehnert, H. Bhatnagar, V. Udandara, S. Albanie, A. Prabhu, and M. Bethge, “A sober look at progress in language model reasoning: Pitfalls and paths to reproducibility,” in *Conference on Language Modeling (COLM)*, 2025.
- [36] Z. Liu, C. Chen, W. Li, P. Qi, T. Pang, C. Du, W. S. Lee, and M. Lin, “Understanding R1-Zero-like training: A critical perspective,” in *Conference on Language Modeling (COLM)*, 2025, arXiv:2503.20783.
- [37] Qwen Team, “Qwen2.5 technical report,” arXiv:2412.15115, 2024.
- [38] Z. Yuan, H. Yuan, C. Li, G. Dong, K. Lu, C. Tan, C. Zhou, and J. Zhou, “Scaling relationship on learning mathematical reasoning with large language models,” arXiv:2308.01825, 2023.
- [39] E. Zelikman, Y. Wu, J. Mu, and N. D. Goodman, “STaR: Bootstrapping reasoning with reasoning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [40] A. Singh, J. D. Co-Reyes, R. Agarwal, A. Anand, P. Patil, X. Garcia, P. J. Liu, J. Harrison, J. Lee, K. Xu *et al.*, “Beyond human data: Scaling self-training for problem-solving with language models,” *Transactions on Machine Learning Research*, 2024.
- [41] A. Havrilla, Y. Du, S. C. Raparthy, C. Nalmpantis, J. Dwivedi-Yu, M. Zhuravinskyi, E. Hambro, S. Sukhbaatar, and R. Raileanu, “Teaching large language models to reason with reinforcement learning,” arXiv:2403.04642, 2024.
- [42] T. Chu, Y. Zhai, J. Yang, S. Tong, S. Xie, D. Schuurmans, Q. V. Le, S. Levine, and Y. Ma, “SFT memorizes, RL generalizes: A comparative study of foundation model post-training,” in *International Conference on Machine Learning (ICML)*, 2025.
- [43] C. Dwork and A. Roth, “The algorithmic foundations of differential privacy,” *Found. Trends Theor. Comput. Sci.*, vol. 9, no. 3–4, p. 211–407, Aug. 2014. [Online]. Available: <https://doi.org/10.1561/04000000042>
- [44] S. Vadhan, “The complexity of differential privacy,” in *Tutorials on the Foundations of Cryptography: Dedicated to Oded Goldreich*, Y. Lindell, Ed. Springer, 2017, pp. 347–450.
- [45] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” arXiv:1707.06347, 2017.
- [46] J. Schulman, “Approximating KL divergence,” <http://joschu.net/blog/kl-approx.html>, 2020, blog post, March 2020.
- [47] Y. Tang and R. Munos, “On a few pitfalls in KL divergence gradient estimation for RL,” arXiv:2506.09477, 2025.
- [48] B. Balle, G. Barthe, and M. Gaboardi, “Privacy amplification by subsampling: Tight analyses via couplings and divergences,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [49] Y. Zhu, J. Dong, and Y.-X. Wang, “Optimal accounting of differential privacy via characteristic function,” in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.
- [50] I. Mironov, “Rényi differential privacy,” in *2017 IEEE 30th Computer Security Foundations Symposium (CSF)*. IEEE, Aug. 2017, p. 263–275. [Online]. Available: <http://dx.doi.org/10.1109/CSF.2017.11>
- [51] I. Mironov, K. Talwar, and L. Zhang, “Rényi differential privacy of the sampled Gaussian mechanism,” arXiv:1908.10530, 2019.
- [52] D. Yu, H. Zhang, W. Chen, J. Yin, and T.-Y. Liu, “Large scale private learning via low-rank reparametrization,” in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, M. Meila and T. Zhang, Eds., vol. 139. PMLR, 18–24 Jul 2021, pp. 12 208–12 218. [Online]. Available: <https://proceedings.mlr.press/v139/yyu21f.html>
- [53] J. Dong, A. Roth, and W. J. Su, “Gaussian differential privacy,” *Journal of the Royal Statistical Society Series B*, vol. 84, no. 1, pp. 3–37, 2022.
- [54] B. Balle, G. Barthe, M. Gaboardi, J. Hsu, and T. Sato, “Hypothesis testing interpretations and Rényi differential privacy,” in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020.
- [55] N. Papernot and T. Steinke, “Hyperparameter tuning with Rényi differential privacy,” in *International Conference on Learning Representations (ICLR)*, 2022.
- [56] I. Mironov, “On significance of the least significant bits for differential privacy,” in *ACM Conference on Computer and Communications Security (CCS)*, 2012, pp. 650–661.
- [57] J. Jin, E. McMurtry, B. I. P. Rubinstein, and O. Ohrimenko, “Are we there yet? Timing and floating-point attacks on differential privacy systems,” in *IEEE Symposium on Security and Privacy (S&P)*, 2022, pp. 473–488.
- [58] Google Differential Privacy Team, “Google differential privacy library: dp_accounting,” <https://github.com/google/differential-privacy>, 2023, Python package dp-accounting, version 0.4.3 (released September 2023).
- [59] B. Balle, L. Berrada, Z. Charles, C. A. Choquette-Choo, S. De, V. Doroshenko, D. Dvijotham, G. Andrew, A. Ganesh, S. Ghalebikesabi, J. Hayes, P. Kairouz, R. McKenna, B. McMahan, A. Pappu, N. Ponomareva, M. Pravilov, K. Rush, S. L. Smith, and R. Stanforth, “JAX-Privacy: Algorithms for privacy-preserving machine learning in JAX,” https://github.com/google-deeppmind/jax_privacy, 2025.
- [60] D. Hendrycks, C. Burns, S. Kadavath, A. Arora, S. Basart, E. Tang, D. Song, and J. Steinhardt, “Measuring mathematical problem solving with the MATH dataset,” in *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*, 2021. [Online]. Available: <https://openreview.net/forum?id=7Bywt2mQsCe>
- [61] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman, “Training verifiers to solve math word problems,” arXiv:2110.14168, 2021.
- [62] A. Talmor, J. Herzig, N. Lourie, and J. Berant, “CommonsenseQA: A question answering challenge targeting commonsense knowledge,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2019, pp. 4149–4158.
- [63] A. Patel, S. Bhattamishra, and N. Goyal, “Are NLP models really able to solve simple math word problems?” in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2021, pp. 2080–2094.
- [64] O. Kallenberg, *Foundations of Modern Probability*, 3rd ed., ser. Probability Theory and Stochastic Modelling. Springer, 2021, vol. 99.
- [65] F. Liese and I. Vajda, “On divergences and informations in statistics and information theory,” *IEEE Transactions on Information Theory*, vol. 52, no. 10, pp. 4394–4412, 2006.
- [66] A. Yousefpour, I. Shilov, A. Sablayrolles, D. Testuggine, K. Prasad, M. Malek, J. Nguyen, S. Ghosh, A. Bharadwaj, J. Zhao, G. Cormode, and I. Mironov, “Opacus: User-friendly differential privacy library in PyTorch,” arXiv:2109.12298; NeurIPS 2021 Workshop on Privacy in Machine Learning, 2021, version 1.5.2.
- [67] S. Gopi, Y. T. Lee, and L. Wutschitz, “Numerical composition of differential privacy,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [68] L. von Werra, Y. Belkada, L. Tunstall, E. Beeching, T. Thrush, N. Lambert, S. Huang, K. Rasul, and Q. Gallouédec, “TRL: Transformers reinforcement learning,” <https://github.com/huggingface/trl>, 2020, version 0.15.2.

APPENDIX A PROOFS FOR SECTION VI

A. Notation

Throughout, $D \simeq D'$ are neighboring prompt datasets in the sense of Definition 3, and d is the number of LoRA coordinates. Because that relation is symmetric and (1) is required on every ordered pair, it suffices to treat pairs in which slot k is present in D and absent in D' and to bound both $H_\alpha(\cdot||\cdot)$ orders;

this is what Lemma 1 does. We add a round index to the quantities of Algorithm 1: $c_{t,i}$ is the coin of line 2, $\bar{g}_{t,i}$ the clipped gradient of line 7, and z_t the noise of line 8. Round t computes the unnormalized release

$$U_t = \sum_{i=1}^n c_{t,i} b_i \bar{g}_{t,i} + z_t, \quad \hat{g}_t = \frac{U_t}{qn}. \quad (9)$$

Since qn is public, $U_t \mapsto \hat{g}_t$ is a bijection with measurable inverse, so by Lemma 2(i) below every hockey-stick divergence between round laws is the same whether computed on U_t or on $\hat{g}_t$; we compute on U_t , where the algebra is simplest, and state every result for the release $\hat{g}_t$, which is the object of Section VI. Write $h_{<t} = \hat{g}_{1:t-1}$ for the prior releases, $\mathcal{L}_D(X)$ for the law of a random variable X when the input is D , and $\mathcal{M}_t(D \mid h_{<t}) = \mathcal{L}_D(\hat{g}_t \mid h_{<t})$ for the conditional law of round t . By (A1), $\theta_t = \Phi_t(h_{<t})$ for a fixed public map Φ_t that iterates line 9 from θ_1 ; if public evaluations influence training, they are part of Φ_t . The final checkpoint is $\theta_{T+1} = \Phi_{T+1}(\hat{g}_{1:T})$.

By (A2), (A3), and a standard randomization (transfer) argument [64], each clipped gradient can be realized as

$$\bar{g}_{t,i} = \text{clip}_C(g(\theta_t, x_i, \omega_{t,i})), \quad \text{clip}_C(v) = v \min\left(1, \frac{C}{\|v\|_2}\right), \quad (10)$$

for a fixed measurable map g and randomness $\omega_{t,i}$ that is independent across slots and rounds and independent of all coins and noise. The map g includes the fixed rule that replaces a non-finite prompt gradient by zero.

B. Hockey-stick divergence and domination

The hockey-stick divergence H_α of Definition 2 is the f -divergence with convex $f(u) = (u - \alpha)_+$. A mechanism $\mathcal{M}$ is (ϵ, δ) -DP if and only if $H_{\epsilon^e}(\mathcal{M}(D) \parallel \mathcal{M}(D')) \leq \delta$ for every ordered neighboring pair [48].

Definition 4 (Domination [49]): An ordered pair of measures (P, Q) dominates (M, M') if $H_\alpha(M \parallel M') \leq H_\alpha(P \parallel Q)$ for all $\alpha \geq 0$.

Throughout, $A = A_{q,\sigma}$ and $B = B_\sigma$ of (5), so Lemma 1 states that (A, B) dominates one round in the remove direction and (B, A) in the add direction. These are tight dominating pairs for the Poisson-subsampled Gaussian mechanism [48], [49], which is why $\delta_{q,\sigma,T}$ of (6) is the quantity the PLD accountant computes. Requiring domination for all $\alpha \geq 0$, not only $\alpha \geq 1$, is what makes it compose (Lemma 4).

Lemma 2 (Properties of H_α): For $\alpha \geq 0$:

- (i) *Data processing.* $H_\alpha(KP \parallel KQ) \leq H_\alpha(P \parallel Q)$ for every Markov kernel K , with equality if K is a bijection with measurable inverse.
- (ii) *Joint convexity.* For measurable kernels $u \mapsto P_u, Q_u$ and a common mixing law μ , $H_\alpha(\int P_u d\mu \parallel \int Q_u d\mu) \leq \int H_\alpha(P_u \parallel Q_u) d\mu \leq \sup_u H_\alpha(P_u \parallel Q_u)$.
- (iii) *Common factors.* $H_\alpha(P \otimes R \parallel Q \otimes R) = H_\alpha(P \parallel Q)$.

Proof: Parts (i) and (ii) hold for every f -divergence with convex f [65]. For (iii), appending an independent draw from R and discarding the second coordinate are both Markov kernels, so (i) gives both inequalities. ■

C. Proof of Lemma 1

Lemma 3 (Conditional decomposition of one round): Assume (A1)–(A3). Let D' be D with occupied slot k removed, and fix $h_{<t}$. There are random variables (S, V, c, Z) , given by the same measurable functions of the round- t randomness under D and under D' and hence with the same joint law under both, such that S , (c, V) , and Z are mutually independent, $c \sim \text{Bernoulli}(q)$ is independent of V , $\|V\|_2 \leq C$, $Z \sim \mathcal{N}(0, \sigma^2 C^2 I_d)$, and the conditional laws of U_t given $h_{<t}$ are

$$\mathcal{L}_D(U_t \mid h_{<t}) = \mathcal{L}(S + cV + Z), \quad \mathcal{L}_{D'}(U_t \mid h_{<t}) = \mathcal{L}(S + Z). \quad (11)$$

Proof: By (A1), $h_{<t}$ fixes θ_t . Set $S = \sum_{i \neq k} c_{t,i} b_i \bar{g}_{t,i}$, $V = \bar{g}_{t,k}$, $c = c_{t,k}$, and $Z = z_t$. For $i \neq k$ the record (b_i, x_i) is the same in D and D' , and by (10) $\bar{g}_{t,i}$ is a function of $(\theta_t, x_i, \omega_{t,i})$; hence S is a function of $(c_{t,-k}, \omega_{t,-k})$ with the same law under both datasets. By (A3) the round- t coins, rollout randomness, and noise are fresh, mutually independent, and independent of $h_{<t}$, so S , (c, V) , and Z are independent and c is independent of V . Clipping gives $\|V\|_2 \leq C$. Under D' , slot k is absent and contributes the zero vector. ■

Proof of Lemma 1: Fix $h_{<t}$ and let D' be D with slot k removed. By Lemma 3, conditioned on $(S, V) = (s, v)$ the two conditional laws of U_t are

$$P_{s,v} = (1 - q)\mathcal{N}(s, \sigma^2 C^2 I_d) + q\mathcal{N}(s + v, \sigma^2 C^2 I_d), \quad (12)$$

$$Q_s = \mathcal{N}(s, \sigma^2 C^2 I_d). \quad (13)$$

Let R be orthogonal with $Rv = \|v\|_2 e_1$ and apply the bijection $y \mapsto R(y - s)/C$; by Lemma 2(i) this preserves H_α . The images are $A_a \otimes \mathcal{N}(0, \sigma^2 I_{d-1})$ and $B \otimes \mathcal{N}(0, \sigma^2 I_{d-1})$, where $A_a = (1 - q)\mathcal{N}(0, \sigma^2) + q\mathcal{N}(a, \sigma^2)$ and $a = \|v\|_2/C \in [0, 1]$. Lemma 2(iii) removes the common factor. The Markov kernel $K_a(y) = ay + \xi$ with independent $\xi \sim \mathcal{N}(0, (1 - a^2)\sigma^2)$ maps $\mathcal{N}(m, \sigma^2)$ to $\mathcal{N}(am, \sigma^2)$, hence A to A_a and B to B . By Lemma 2(i),

$$H_\alpha(P_{s,v} \parallel Q_s) \leq H_\alpha(A \parallel B), \quad H_\alpha(Q_s \parallel P_{s,v}) \leq H_\alpha(B \parallel A). \quad (14)$$

Finally, $\mathcal{L}_D(U_t \mid h_{<t}) = \int P_{s,v} d\mu$ and $\mathcal{L}_{D'}(U_t \mid h_{<t}) = \int Q_s d\mu$ for the common law μ of (S, V) , so Lemma 2(ii) gives domination of the pair of conditional laws of U_t by (A, B) and of the reversed pair by (B, A) . Applying the bijection $U_t \mapsto \hat{g}_t = U_t/(qn)$ (Lemma 2(i)) transfers both bounds to $(\mathcal{M}_t(D \mid h_{<t}), \mathcal{M}_t(D' \mid h_{<t}))$, which is (8). Nothing in the argument depends on t , $h_{<t}$, G , or C . ■

D. Composition step of Theorem 1

The proof of Theorem 1 in Section VI invokes one external result, stated here in the notation of this appendix.

Lemma 4 (Adaptive composition [49]): Fix an ordered pair (D, D') and measurable kernels $h_{<t} \mapsto \mathcal{M}_t(\cdot \mid h_{<t})$. If for every t and every $h_{<t}$ the pair (P_t, Q_t) dominates $(\mathcal{M}_t(D \mid h_{<t}), \mathcal{M}_t(D' \mid h_{<t}))$, then $(\bigotimes_t P_t, \bigotimes_t Q_t)$ dominates $(\mathcal{L}_D(\hat{g}_{1:T}), \mathcal{L}_{D'}(\hat{g}_{1:T}))$.

The tradeoff-function composition theorem of Gaussian differential privacy [53] gives the same conclusion. Applied with

$(P_t, Q_t) = (A, B)$ for the remove direction and (B, A) for the add direction, which Lemma 1 supplies uniformly in $h_{<t}$, it yields

$$H_{e^\epsilon}(\mathcal{L}_D(\hat{g}_{1:T}) \parallel \mathcal{L}_{D'}(\hat{g}_{1:T})) \leq H_{e^\epsilon}(A^{\otimes T} \parallel B^{\otimes T}), \quad (15)$$

$$H_{e^\epsilon}(\mathcal{L}_{D'}(\hat{g}_{1:T}) \parallel \mathcal{L}_D(\hat{g}_{1:T})) \leq H_{e^\epsilon}(B^{\otimes T} \parallel A^{\otimes T}), \quad (16)$$

both at most $\delta_{q,\sigma,T}(\epsilon)$ of (6); this is the step used in the proof of Theorem 1.

E. Corollaries

Corollary 1 (Replacing a prompt): Under the assumptions of Theorem 1, let $\delta = \delta_{q,\sigma,T}(\epsilon)$. The output $(\theta_2, \dots, \theta_{T+1})$ satisfies (1) with $(2\epsilon, (1 + e^\epsilon)\delta)$ for every pair of prompt datasets that differ only in the record at one occupied slot.

Proof: Let D and D' differ by the prompt at one occupied slot, and let D_0 be either dataset with that slot absent. $D \simeq D_0 \simeq D'$, so applying Theorem 1 twice gives, for every output set O , $\Pr[O \mid D] \leq e^\epsilon \Pr[O \mid D_0] + \delta \leq e^{2\epsilon} \Pr[O \mid D'] + (1 + e^\epsilon)\delta$. Accounting directly under the replace-one relation can be tighter. ■

Corollary 2 (Per-record variants of lines 4–6): Let Algorithm 1' be Algorithm 1 with lines 4–6 replaced by any computation that, given θ_t , maps the record x_i and fresh private randomness to a gradient g_i . Under (A1)–(A4), Algorithm 1' satisfies Theorem 1 with the same (q, σ, T) . In particular this covers verifier-filtered DP-SFT, which samples G traces from the public π_{ref} , keeps those the verifier accepts, and sets g_i to the gradient of their mean token-normalized log-likelihood (zero if none is accepted), and gold-SFT, which sets g_i to the gradient of the log-likelihood of the reference solution and takes the problem together with that solution as the record.

Proof: The realization (10) places no restriction on the map g beyond measurability, and Lemmas 3, 1, and 4 use lines 4–6 only through it. Both comparators keep lines 2–3 and 7–9, so (9) and (10) describe them with a different g , and the proof of Theorem 1 goes through unchanged. ■

Remark 1 (Repeated releases): The guarantee is per run. Releasing several runs trained on the same records composes their transcripts, for example by composing their privacy-loss distributions.

F. Numerical accounting of every configuration

Table VI lists every (n, m, σ, T) used in this paper with $\delta = 1/(10n)$, the certified PLD budget, and the corresponding Rényi-DP (RDP) budget. The PLD column is the `dp_accounting` 0.4.3 accountant with the add-or-remove relation, which composes the privacy-loss distributions of (A, B) and (B, A) and reports the larger ϵ ; its default pessimistic connect-the-dots discretization upper-bounds the exact hockey-stick curve [28]. Since $\delta_{q,\sigma,T}$ is nonincreasing in ϵ , a returned value ϵ^* certifies (ϵ, δ) -DP for every $\epsilon \geq \epsilon^*$ by Theorem 1. Configurations that share (n, m, σ, T) share both budgets regardless of loss, clipping norm, learning rate, or verifier (Corollary 2); in particular the DP-SFT and gold-SFT comparators are accounted identically to DP-GRPO.

TABLE VI
EVERY ACCOUNTING CONFIGURATION IN THIS PAPER, WITH $\delta = 1/(10n)$
AND $q = m/n$, UNDER PLD (REPORTED) AND RDP ACCOUNTING.
PARENTHEZIZED: THE MINIMIZING RDP ORDER.

Configuration	n	(m, σ, T)	PLD	RDP
MATH, $\epsilon=1$ arm	6,988	(128, 1.002, 50)	1.000	1.463 (7.6)
MATH, $\epsilon=4.23$ arm	6,988	(128, 0.612, 50)	4.235	5.322 (3.2)
MATH, reference [†]	6,988	(128, 0.484, 50)	8.000	9.830 (2.3)
MATH, DP-SFT $T=100$	6,988	(128, 0.519, 100)	8.000	9.702 (2.3)
MATH, coverage	6,988	(512, 0.702, 50)	8.000	9.323 (2.6)
MATH, hist. rank-4	6,988	(128, 0.612, 100)	5.038	6.185 (3.0)
GSM8K, CSQA [‡]	6,961	(128, 0.520, 100)	8.000	9.701 (2.3)

[†]Shared by the zero-advantage control, clipping-norm study, lower-compute arm, both DP-SFT $T=50$ pairs, and gold-SFT. [‡]Every arm on both benchmarks. MATH, hist. rank-4 is the canary run (Appendix B).

The RDP column composes the RDP of the Poisson-subsampled Gaussian mechanism, evaluated exactly at integer orders and by a convergent series at fractional orders [51], over T rounds and converts to (ϵ, δ) with the improved conversion of Balle et al. [54], minimized over the default order grid of `dp_accounting` 0.4.3 ($\{1.1, 1.2, \dots, 10.9\} \cup \{11, \dots, 63\} \cup \{128, 256, 512, 1024\}$). Opacus [66] applies the same conversion over a slightly smaller default grid ($\{1.1, \dots, 10.9\} \cup \{12, \dots, 63\}$) and returns the same values as Table VI to four decimal places. The two accountants bound the same composition; the PLD value is the tighter one, and the gap comes almost entirely from the RDP-to- (ϵ, δ) conversion, which is loose at large ϵ and small σ [49], [54], [67]. A reader who prefers RDP should read the $\epsilon \in \{1, 4.23, 8\}$ frontier as $\epsilon \in \{1.46, 5.32, 9.83\}$.

G. Status of the assumptions

a) (A1): Holds for every declared run: each run's public inputs are fixed before it starts, the step count T is fixed in advance, line 8 divides by the public qn rather than the realized batch size, and constant-reward groups are retained, so no choice introduces data-dependent control flow. Hyperparameters were selected on pilot runs over the same training prompts, following the convention of the private fine-tuning literature [11], [24], and, as in that literature, the stated budget covers the reported run and not the selection; a private tuning analysis [55] would add a bounded amount. One supervised comparator, the resumed 100-update DP-SFT run, had its verifier-trip limit raised after a trip on private completions (Appendix C-E0b); that choice is the one run-time decision made with access to D , and it concerns a comparator, not DP-GRPO.

b) (A2): Enforced on the training side: each prompt group is backpropagated separately (line 6), clipping follows chunk accumulation (line 7), and a non-finite prompt gradient is replaced by zero by a fixed per-record rule. The cross-prompt isolation and chunk-identity tests in Appendix B check this property. A non-finite *noisy* update skips the optimizer step, which is post-processing. Two execution effects can break (A2) outside exact, isolated computation, and the theorem does not cover them. First, the rollouts of line 4 for all sampled

prompts are submitted as independent single-sequence requests to one inference engine. In exact arithmetic, batch composition does not change any request’s conditional token distribution; floating-point inference kernels are not guaranteed to be batch-invariant, so a prompt’s realized rollouts may depend on which prompts share its batch. Second, the bounded training verifier of line 5 applies a 10-second wall-clock limit, so a reward can depend on machine load created by other records. We measure neither effect. Verifier-filtered DP-SFT (Corollary 2) generates its traces through the same engine and filters them with the same verifier, so both gaps apply to it; gold-SFT generates nothing and calls no verifier, so neither applies.

c) (A3): Holds for every released model. Their sampling and noise generators are seeded from operating-system entropy, and recovery files, including generator states and verifier trip logs that contain private completions, are not released. The earlier design-study runs seeded their generators from documented experiment seeds for reproducibility (Appendix C-A); they are utility measurements and are not among the released models.

d) (A4): Idealized. Floating-point noise generation can violate DP guarantees that hold over the reals [56], [57], and the measured clipped sensitivity exceeds C by one unit in the last place. Data-dependent system telemetry, such as per-step overflow counts, is outside the release covered here. The theorem certifies the mechanism, not its finite-precision execution.

APPENDIX B MECHANISM VERIFICATION

The tests in this appendix verify that the implementation realizes the stated mechanism. They do not replace the differential-privacy proof or accountant. The machine-readable record is `results/mechanism_verification.json`; it records Python, PyTorch, `dp-accounting`, Opacus, and NumPy versions with a UTC timestamp.

a) Global add/remove sensitivity.: We form a base dataset of five prompt gradients for a two-layer double-precision model. For each of 20 random additions and one adversarial addition with magnitude 10^{12} , we recompute the sum of independently clipped prompt gradients. With $C = 0.7$, the largest observed $\|f(D') - f(D)\|_2 / C$ over 21 trials exceeds one by 2.2×10^{-16} , one binary64 unit in the last place, below the test tolerance 10^{-10} .

b) Non-private equivalence.: At $q = 1$, $\sigma = 0$, and $C = \infty$, the trainer averages n prompt losses, each of which averages exactly G rollout losses. For equal group size, this is the flat mean over all nG sequences. A double-precision SGD step through the private trainer and a flat baseline has maximum absolute parameter difference zero at tolerance 10^{-12} .

c) Poisson frequency and independence.: For six public slots sampled independently with $q = 0.2$ over 20,000 trials, the maximum absolute inclusion-frequency error is 0.0040 and the maximum absolute off-diagonal inclusion correlation is 0.0141. These are below the preregistered tolerances 0.012 and

0.035, respectively. This CPU gate is separate from the rollout-generation regression test, which requires each completion to be an unseeded single-sequence request.

d) Cross-prompt isolation.: The test computes prompt i ’s gradient, evaluates prompt j , changes all of prompt j ’s rewards, evaluates it again, and then recomputes prompt i . Prompt i ’s gradient is bit-identical before and after the perturbation. A second test exercises this property through the production adapter and trainer, so batching cannot move reward statistics across prompt boundaries.

e) Zero-advantage groups.: All-correct and all-wrong groups produce exactly zero normalized advantages. Both losses and gradients remain finite; the two gradients are bit-identical and nonzero because the KL term remains active. Such records are retained and clipped rather than removed through data-dependent control flow.

f) External accountant validation.: For $q = 0.1$, noise multiplier 3.0, 512 steps, and $\delta = 10^{-6}$, the repository privacy-loss-distribution (PLD) accountant returns $\epsilon = 3.7697$, against the published JAX-Privacy reference of 3.77 (absolute difference 3.1×10^{-4}). Opacus RDP returns the larger value 4.052 (order 7).

g) Rollout-chunked backward identity.: For eight rollouts, the test compares one unchunked prompt backward pass with four chunked passes of size two accumulated into the same prompt gradient. The detached scalar losses agree at $(\text{rtol}, \text{atol}) = (10^{-6}, 10^{-7})$, and every parameter gradient agrees at $(10^{-10}, 10^{-12})$. The production-adapter test repeats the comparison for chunk sizes one, two, four, and eight. Clipping occurs only after all chunks have accumulated, so chunking changes memory geometry but not the prompt record or clipping boundary.

h) Separate fp16 preflight.: At each of the three clipping norms, 21 sensitivity trials give a maximum ratio one fp32 ULP (1.2×10^{-7}) above one. Cross-prompt gradients are bit-identical and constant-reward groups are retained. The designated chunk-identity case has maximum absolute difference 2^{-12} , so it is not bit-identical. A follow-up set records no clip-decision flips at the three tested norms; neither test bounds the discrepancy near every clipping threshold.

i) Reproduction entry points.: `scripts/emit_verification_artifact.py` produces the machine-readable record from the suites `tests/test_dp_trainer.py`, `tests/test_chunked_grpo.py` and `tests/test_math_dp_adapter.py`; the fp16 records are kept separately from the binary64 ones.

j) Canary diagnostic.: One 100-update bf16 run in the canary configuration of Table VII (nominal $\epsilon = 5.04$ at $\delta = 1/69,880$) replaced 1,000 of the 6,988 training slots with randomized canaries, 532 present and 468 null. With $q = 128/6,988$, a present canary is sampled 1.83 times in expectation and never with probability $(1 - q)^{100} = 0.157$; realized inclusions were not logged. The empirical lower bound returned is 0.0. The audit therefore gives no positive lower bound and says nothing about the tightness of the accountant; its generators were seeded for reproducibility, so it is a

mechanism diagnostic rather than one of the released models (Appendix C-A). A powered audit is future work (Section IX).

APPENDIX C CONFIGURATIONS AND PROVENANCE

A. Shared Training Configuration

Every arm trains rank-4 LoRA adapters ($\alpha=8$, dropout 0.05) on the seven attention and MLP projection matrices of Qwen2.5-1.5B-Instruct, $p = 4,616,192$ trainable coordinates in 392 tensors, with AdamW; the noise $z_t/(qn)$ of Algorithm 1 has expected norm $\sigma C \sqrt{p}/(qn) \approx 0.45$ on MATH and 0.48 on GSM8K, eight to nine times the largest possible averaged clipped signal $C = 0.055$. Training runs on one RTX 6000 GPU with fp16 weights and fp32 loss, clipping, noise and optimizer arithmetic. GRPO completions are drawn as independent, unseeded single-sequence requests. Poisson sampling and Gaussian noise draw their randomness from operating-system entropy, independently of the initialization seed. The earlier design-study runs (the budget sweep of Table XII, the non-private references in Tables III and IV, and the canary run of Appendix B) were trained in bf16 on different hardware and kernels with generators seeded for reproducibility; each is compared only with the base evaluated under its own precision, and the two families are never pooled.

Training uses the GRPO trainer of TRL version 0.15.2 [68] with one optimizer step per rollout batch. The loss (7), the advantage normalization (2) with $c = 10^{-4}$ and a $G - 1$ standard-deviation denominator, and the per-token KL estimator (3) reproduce that trainer’s computation; the per-prompt clipping, noise, and qn normalization of Algorithm 1 replace its batch-level gradient step.

B. Mechanism Parameters

Table VII gives every mechanism configuration at full precision; the body rounds these to three decimals. The sampling rate is $q = m/n$ exactly and $\delta = 1/(10n)$, i.e. $1/69,880$ on MATH and $1/69,610$ on GSM8K and CommonsenseQA. The clipping norm is $C = 0.05536$ for DP-GRPO and the zero-advantage control, with $C \in \{0.02768, 0.11072\}$ in the clipping sweep, and $C = 0.1143670583$ for both supervised arms, inherited from the private fine-tuning recipe they follow [11], [24] and not swept; their learning rates are 5×10^{-6} (filtered) and 2×10^{-5} (gold), selected in Appendix C-D. Table VI gives the resulting budgets.

C. Evaluation Sets and Scoring

a) *MATH*.: The 4,499-problem endpoint is the 5,000-problem test split minus one problem duplicated in the training set and the 500 MATH-500 problems that earlier runs, and the supervised learning-rate pilots (Appendix C-D), used for selection. The exclusion is by problem identity from saved 4,999-problem predictions; no model is rerun. It was fixed after those earlier runs. The base model solves 2,207 problems (49.1%) under fp16 and 2,212 (49.2%) under bf16; each study is compared with the base of its own precision, and neither correctness vector is substituted for the other.

TABLE VII
MECHANISM PARAMETERS AT FULL PRECISION. m IS THE EXPECTED NUMBER OF PROMPTS PER ROUND.

Configuration	n	m	T	σ
MATH, reference [†]	6,988	128	50	0.4842404738068581
MATH, $\epsilon=4.23$ (bf16)	6,988	128	50	0.612058477104
MATH, $\epsilon=1$ (bf16)	6,988	128	50	1.0023417361080647
MATH, coverage ($G=2$)	6,988	512	50	0.7022244453430175
MATH, DP-SFT, 100 updates	6,988	128	100	0.519267738191411
MATH, canary (bf16)	6,988	128	100	0.612058477104
GSM8K and CSQA, all arms	6,961	128	100	0.5196564674377442

[†] Shared by DP-GRPO and the control, the $\epsilon=8$ budget-sweep cell, the clipping sweep, the lower-compute arm ($G=2$), and the $T=50$ DP-SFT arms.

b) *GSM8K, CommonsenseQA and SVAMP*.: GSM8K is scored on its 1,319 test problems. The 6,961 CommonsenseQA training questions are drawn uniformly at random from the training split after excluding 700 questions used in a pilot, and all 1,221 validation questions are scored. SVAMP is scored with GSM8K’s prompt, decoding and cap (Appendix D-D).

c) *Statistics*.: Problem-cluster bootstrap intervals use 100,000 draws and sign-randomization p -values use 1,000,000 draws, each from a fixed seed recorded with the analysis. On CommonsenseQA, Holm adjustment runs across its eight contrasts (four comparators under two scorings).

D. Learning-Rate Pilots

Each supervised recipe’s learning rate was chosen by one pilot run per value in $\{5 \times 10^{-6}, 2 \times 10^{-5}, 8 \times 10^{-5}, 3 \times 10^{-4}\}$ on MATH, with one initialization seed and the private configuration of Table VII, selected by MATH-500 accuracy at step 50 with ties broken toward the smaller rate. MATH-500 is disjoint from the 4,499-problem endpoint. Filtered DP-SFT selected 5×10^{-6} and gold DP-SFT 2×10^{-5} ; both were carried unchanged to GSM8K and CommonsenseQA. In increasing order of rate, the filtered pilots scored 50.6, 49.2, 49.6 and 50.6% on MATH-500, a flat profile whose selection is a tie, and the gold pilots scored 47.8, 48.2, 40.6 and 29.4%. The public base answers 48.6% of the same problems under the endpoint evaluation, so no pilot rate improves on it by more than two points. Training loss is private telemetry and was not logged (Section IV), so these pilots are the only evidence on the supervised learning rates. As is conventional in private fine-tuning, the pilots’ use of the training prompts is not charged to the reported budget (Appendix A-G).

E. Resumed and Recovered Runs

Six reported trajectories were resumed from checkpoints. Each checkpoint stores the optimizer, privacy ledger, and framework, CUDA, sampling and noise generator states, which are restored after inference-engine initialization. None is claimed to be bitwise equivalent to uninterrupted execution, and the recovery states are not part of any release.

a) *MATH clipping sweep, high- C seed 2*.: The $C = 0.11072$ seed-2 run resumed from checkpoint 35, replaying updates 36–40 and then continuing to 50: 55 optimizer computations, 50 compositions in its final ledger. The inference

TABLE VIII
PROTOCOL DECISIONS AND THEIR TIMING.

Decision	Fixed when
Strict scorers (all benchmarks)	Before any fixed-protocol arm was run (Sec. VII-D).
DP-GRPO LR, C , G , T on MATH	In the MATH design study, before the fixed protocol (Sec. VII-B).
Supervised learning rates	Four-point pilots on MATH-500, before the fixed protocol (App. C-D).
GSM8K horizon and cap	Before any private GSM8K run, from one earlier non-private run (Sec. VII-A).
MATH-500 excluded from evaluation	After the earlier runs that used it for selection; applied by problem identity to saved predictions (App. C-C).
Seeds 3–4, DP-GRPO and control, MATH and GSM8K	After the seed 1–2 results were inspected (Sec. VII-C); four-seed contrasts are sequential.
CSQA answer declaration format	After a pilot with a stricter marker failed, before the CSQA arms (Sec. VII-A).
CSQA final-answer rescore rule	After strict results of base, DP-GRPO, control and gold SFT were known; hash-frozen before rescoring them and before filtered-SFT evaluation (App. E-B).
MATH format-robust rules	Developed on public-base predictions; two arm-blind audits before arm-level rescoring (App. E).
Verifier trip limit $5 \rightarrow 10$	After one 100-update DP-SFT run tripped; that run resumed (App. C-E0b).
Resumed trajectories	High- C seed 2 (updates 36–40 replayed); 100-update DP-SFT seed 1 (46–50 replayed); CSQA seeds 1–2 of DP-GRPO and control (no replay); three allocation runs admitted at their final checkpoint (App. C-E).

engine reseeded framework and CUDA randomness after the dedicated generators were restored. Table XII reports this trajectory, not its clean replication. The ledger accounts for the retained trajectory only; a release that included the discarded branch as well would have to compose both.

b) MATH 100-update DP-SFT, seed 1.: The run exceeded the training-verifier trip limit, which was raised from 5 to 10, and resumed from checkpoint 45, replaying updates 46–50: 105 optimizer computations produced the retained 100-update trajectory. The accountant covers the retained trajectory, not the abandoned branch. The trip occurred on private completions and the limit was raised afterwards, so this comparator run depends on one choice made with access to D (Appendix A-G).

c) CommonsenseQA, seeds 1–2 of DP-GRPO and the control.: The four runs paused cleanly before scheduled cluster maintenance, at checkpoints 60–66, and resumed from those checkpoints. Each saved step equalled the number of completed updates, so no update was replayed.

d) Allocation study.: Three allocation runs failed during process shutdown after training had finished. Their final checkpoints were admitted after independent verification, without further training.

F. Protocol Timeline

Table VIII collects every decision that was made after some data or result had been seen, with the runs it preceded and followed; each is discussed where it arises in the text.

TABLE IX
EVERY ARM, THE BENCHMARKS IT RAN ON, ITS SEEDS, AND WHERE ITS RESULTS ARE REPORTED. SEEDS FOR STAGE 2 ARE GIVEN AS MATH / GSM8K / CSQA.

Arm	Benchmark	Seeds	Reported
<i>Stage 1: design study</i>			
$\epsilon \in \{1, 4.23, 8\}$ (bf16)	MATH	2 each	Tab. XII
$C \in \{\frac{1}{2}, 1, 2\} \times 0.055$	MATH	2 each	Tab. XII
Rollout allocation (2 alt.)	MATH	2 each	Tab. XII
Same-LR and 100-update SFT	MATH	2 each	Sec. VIII-B
$\sigma=0$, clipped / unclipped	GSM8K	2 each	Fig. 3
$\sigma=0$, clipped (bf16)	MATH	2	App. D-B
$\sigma=0$, unclipped, LR 10^{-4}	GSM8K	1	App. D-B
<i>Stage 2: fixed protocol</i>			
Public base	all three	–	Fig. 2,
DP-GRPO ($\epsilon=8$)	all three	4 / 4 / 2	Tab. II,
Zero-advantage control	all three	4 / 4 / 2	III, V
Filtered DP-SFT	all three	2 / 2 / 2	
Gold DP-SFT	all three	2 / 2 / 2	
<i>Evaluation only</i>			
GSM8K checkpoints, all arms	SVAMP	as GSM8K	App. D-D
<i>Historical non-private runs (bf16, rank 16)</i>			
Standard GRPO, 900 updates	MATH	1	Tab. III
Standard GRPO, 2 epochs	GSM8K	1	Tab. IV

APPENDIX D SUPPLEMENTARY RESULTS

This appendix gives per-seed results and the supporting design studies. Each contrast is computed within its own study; historical bf16 runs and the fp16 comparisons are kept separate.

A. Per-Seed Results of the Fixed Protocol

TABLE X
STRICT ACCURACY (%) OF EVERY SEED PLOTTED IN FIG. 2, AND THE MATH FORMAT-ROBUST ACCURACIES BEHIND TABLE III. SEEDS IN RUN ORDER.

Arm	Seeds	Mean
<i>MATH, strict</i>		
Public base	–	49.1
Zero-adv. control	48.9 / 48.8 / 49.1 / 48.8	48.9
DP-GRPO	51.3 / 51.9 / 51.8 / 51.3	51.6
Filtered DP-SFT	49.3 / 49.3	49.3
Gold DP-SFT	48.7 / 48.6	48.6
<i>GSM8K, strict</i>		
Public base	–	70.7
Zero-adv. control	70.9 / 71.1 / 71.6 / 70.7	71.1
DP-GRPO	74.1 / 73.9 / 74.3 / 75.0	74.3
Filtered DP-SFT	70.4 / 70.7	70.5
Gold DP-SFT	70.1 / 70.4	70.2
<i>MATH, format-robust</i>		
Public base	–	50.4
Zero-adv. control	50.3 / 50.2 / 50.3 / 50.4	50.3
DP-GRPO	51.3 / 52.0 / 51.8 / 51.3	51.6

B. Budget, Clipping, and Non-Private References

Within the tested ranges, neither the accounted budget nor the clipping norm moves MATH accuracy detectably (Table XII). Relative to $\epsilon=8$, the bf16 budget sweep gives -0.09 [-0.66 ,

TABLE XI

PER-SEED DIFFERENCES (POINTS) BEHIND TABLE II, WITH HOLM-ADJUSTED EXACT MCNEMAR p (HOLM WITHIN EACH CONTRAST). THE SUPERVISED CONTRASTS PAIR DP-GRPO SEEDS 1–2 WITH THE SUPERVISED SEEDS.

Contrast	Per-seed difference	Holm p , largest
<i>MATH</i>		
DP-GRPO – control	+2.31 / +3.16 / +2.62 / +2.51	6.4×10^{-5}
DP-GRPO – filtered	+1.98 / +2.67	2.4×10^{-4}
DP-GRPO – gold	+2.60 / +3.38	2.9×10^{-6}
<i>GSM8K</i>		
DP-GRPO – control	+3.26 / +2.81 / +2.65 / +4.25	0.013
DP-GRPO – filtered	+3.79 / +3.26	1.1×10^{-3}
DP-GRPO – gold	+4.09 / +3.49	7.5×10^{-4}

MATH format-robust DP-GRPO – control per seed: +0.98 / +1.84 / +1.56 / +0.98.

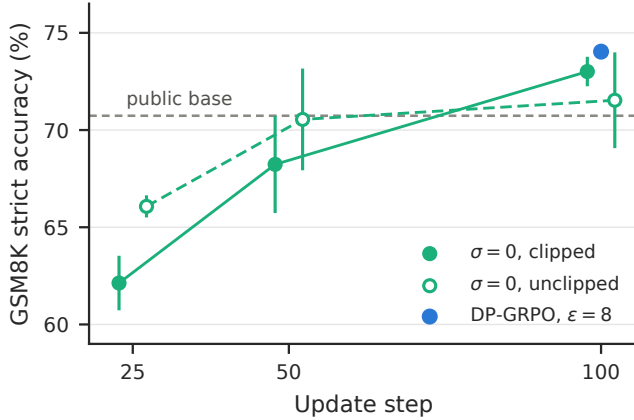


Fig. 3. Non-private GSM8K references at DP-GRPO’s hyperparameters: noise removed, with clipping retained or disabled. Dots show two-seed means; bars span the seeds, not confidence intervals. DP-GRPO is the mean of its two matched seeds at the final step (74.0%). The earlier non-private run of Table IV (75.4%) was trained at a different rank, learning rate and horizon and is not shown.

0.48] at $\epsilon=4.23$ and -0.57 [$-1.19, 0.06$] at $\epsilon=1$, with MDEs 0.82 and 0.89; the interval for an eightfold tighter budget excludes a cost above 1.19 pp. In the fp16 clipping sweep, halving C changes accuracy by -0.30 [$-0.72, 0.12$] and doubling it by -0.03 [$-0.37, 0.30$] (MDEs 0.60 and 0.48). Because the noise scales with C , this sweep moves clipping bias and absolute noise together and does not separate them.

On the mathematical tasks the $\sigma=0$ references at the private hyperparameters do not exceed DP-GRPO (Fig. 3). On GSM8K the matched two DP-GRPO seeds gain +3.30 [1.44, 5.16] over base; the clipped reference gains +2.27 [0.23, 4.32] and the unclipped one +0.80 [$-1.36, 2.96$], its seeds ending at 74.0% and 69.1%. On MATH the two-seed clip-only reference at learning rate 10^{-4} reaches 51.3%, against 51.6% for DP-GRPO, and a single-seed unclipped GSM8K reference at the same rate reaches 74.3% by step 25 and 60.1% at step 100. These references keep hyperparameters selected under noise and are not a non-private ceiling; the retention ratios of Section VIII-E use the earlier standard GRPO runs, and on CommonsenseQA,

TABLE XII
SUPPORTING MATH DESIGN COMPARISONS.

Alternative	Change (pp), 95% CI	MDE (pp)
<i>Budget; reference: $\epsilon = 8$ (earlier study)</i>		
$\epsilon = 4.23$	-0.09 [$-0.66, 0.48$]	0.82
$\epsilon = 1$	-0.57 [$-1.19, 0.06$]	0.89
<i>Clipping; reference: $C = 0.055$</i>		
$C = 0.111$	-0.03 [$-0.37, 0.30$]	0.48
$C = 0.028$	-0.30 [$-0.72, 0.12$]	0.60
<i>Rollouts $\times$ prompts; reference: 8×128</i>		
2×512 (same rollouts)	-0.98 [$-1.73, -0.23$]	1.08
2×128 (quarter rollouts)	-0.96 [$-1.74, -0.18$]	1.12
<i>SFT; reference: LR $5 \cdot 10^{-6}$, 50 updates</i>		
LR $3 \cdot 10^{-4}$ (DP-GRPO’s)	$+0.22$ [$-0.70, 1.14$]	1.32
100 updates	$+0.54$ [$0.03, 1.06$]	0.73

Change is alternative minus the reference named in each block. Two seeds per setting; studies are analyzed separately.

where the reference at the private hyperparameters exceeds DP-GRPO by about six points, the comparison is paired.

Table XII collects these MATH design comparisons with the allocation study and the two supervised challenges. The budget rows are bf16 runs and are compared only with each other. In the SFT block each change is relative to selected-LR SFT, not to DP-GRPO. The high- C clipping pair and the 100-update SFT arm each include one resumed seed (Appendix C-E).

C. Rollouts per Prompt versus Prompts per Round

On MATH, using two rather than eight rollouts per prompt reduces measured training time from 12.1–12.2 hours to 3.7–3.8 hours, a $3.2\times$ reduction, while lowering strict accuracy by 0.96 [0.18, 1.74] points. Reallocating the same expected 1,024 rollouts to 512 prompts with two responses each instead lowers accuracy by 0.98 [0.23, 1.73] (Table XII). The latter changes the sampling rate and its calibrated noise multiplier as well as group size.

Under format-robust scoring, the reference-minus-alternative gaps shrink to +0.49 [$-0.27, 1.26$] for lower compute and +0.62 [$-0.11, 1.37$] for broader coverage. The measured speedup remains, and about half of the strict-accuracy cost is answer formatting rather than correctness, consistent with fewer within-group contrasts per prompt leaving less signal about the output format.

D. Transfer to SVAMP

The released GSM8K checkpoints (step 100, no selection) were scored on the 1,000 problems of SVAMP [63] with GSM8K’s prompt, decoding and 512-token cap, and a numeric extractor frozen before scoring. No SVAMP problem matches a GSM8K training question exactly or with token Jaccard similarity of at least 0.8. Scoring a released model on public data is post-processing, so this costs no privacy budget. Table XIII gives the results. The four DP-GRPO seeds average 83.2%, versus 82.0% for base. All four point estimates are higher; the pooled gap is +1.20 [$-0.53, 2.95$], with Holm-adjusted $p = 0.575$. With 1,000 problems the MDE is 2.49

points, so this check can detect a transfer gain of GSM8K’s size (3.6 points) but not one of a third of that size, which is what the point estimate shows. The seed-level interval is [0.50, 1.90]. Differences from filtered and gold DP-SFT lie within the same MDE. SVAMP is a transfer check, not an additional training benchmark. The six contrasts were Holm-corrected together, including two against the non-private $\sigma=0$ references (77.9% clipped, 78.1% unclipped), which are not in the table: 8.5–12.3% of their outputs reach the length cap, against at most 0.8% for every arm shown, so their SVAMP accuracy reflects response length rather than transfer.

TABLE XIII
SVAMP (1,000 PROBLEMS): GSM8K CHECKPOINTS SCORED WITHOUT FURTHER TRAINING. STRICT ACCURACY PER SEED, AND DP-GRPO’S DIFFERENCE FROM EACH ARM (POINTS, 95% CI).

Arm	Seeds	Mean	DP-GRPO – arm
Public base	–	82.0	+1.20 [−0.53, 2.95]
Zero-adv. control	82.1 / 81.8 / 82.1 / 82.1	82.0	+1.18 [−0.35, 2.73]
Filtered DP-SFT	82.6 / 82.6	82.6	+0.25 [−1.45, 1.95]
Gold DP-SFT	82.2 / 82.7	82.5	+0.40 [−1.40, 2.20]
DP-GRPO	83.0 / 82.7 / 83.4 / 83.7	83.2	–

MDEs 2.49, 2.20, 2.42 and 2.58 points; Holm-adjusted $p \geq 0.58$. The supervised rows use DP-GRPO seeds 1–2.

APPENDIX E

ANSWER EXTRACTION AND SCORING SENSITIVITY

A. MATH: Strict and Format-Robust Scores

Strict accuracy is the fraction of all 4,499 evaluation problems with a correct boxed answer. Format-robust accuracy uses the same denominator and the same saved responses. It preserves scores for boxed answers and adds a fallback for unboxed responses shorter than the 1,024-token cap. Thus a capped response is excluded from the fallback, not from evaluation.

The fallback uses the final explicit answer statement, then a final inferential statement, then the final nonempty line. Within that segment it extracts a mathematical expression, the right-hand side of an assignment, or a stated answer value. It preserves final answers joined by “and” or “or” rather than selecting one. The extractor never reads the gold answer; the extracted candidate is compared with gold by the existing symbolic equality verifier. The rules were developed on public-base predictions. Two arm-blind audits led to corrections before unblinding: excluding capped responses from fallback and preserving multiple answers. The final deduplicated audit found 60/60 candidates consistent with the completed final statement and no observed false positives. No verifier guard tripped across the 47 unique prediction artifacts rescored.

a) *Decomposing the improvement.*: Table XIV partitions problems by the base model’s response. “Questions” is a count. Each contribution is the net number of additional correct answers in that group, averaged over four seeds, divided by all 4,499 problems and multiplied by 100. It is measured in percentage points, not the group’s own accuracy. The contributions therefore sum to the total gain over base, up to

TABLE XIV
CONTRIBUTIONS TO THE MATH GAIN OVER BASE.

Base response	Questions	Contribution (pp)	
		Strict	Format-robust
Boxed an answer	3,831	−0.24	−0.24
No boxed answer	668	+2.75	+1.49
All problems	4,499	+2.51	+1.26

rounding. Problems whose base response has no boxed answer account for +2.75 points under strict scoring but +1.49 under format-robust scoring. The total gains are +2.51 [1.54, 3.48] and +1.26 [0.33, 2.17], respectively.

b) *GSM8K.*: The primary extractor already accepts an unboxed final number. A separate sensitivity analysis counts every response reaching the length cap as incorrect; the reward contrast remains +2.96 [1.33, 4.59] points.

B. CommonsenseQA: Final-Answer Rescoring

The strict score uses the training reward’s letter extractor. Option discussion can cause that extractor to reject a response even when it later states a clear answer. The final-answer rule instead reads the last explicit answer declaration and rejects conflicting choices within that declaration. Both rules use all 1,221 validation questions and the same completions. Rescoring changes neither training rewards nor model weights.

The rule was written after the strict results for base, DP-GRPO, the zero-advantage control, and gold SFT were known. It was hash-frozen before rescoring those arms and before filtered-SFT evaluation. This is a post-hoc sensitivity endpoint. The initial audit of 40 changed extractions found no disagreements and no false positives among 21 rescored-correct responses. A separate filtered-SFT audit sampled 40 of 578 changed records, covering 37 distinct questions/completions. It found no disagreements and no false positives among 23 rescored-correct responses. Seeds, gold answers, and extracted answers were hidden in that audit, but the filtered-SFT arm identity was known. These were agent audits, not independent human assessments; zero observed errors does not establish an error-free extractor.

The base has 815 correct answers and 21 remaining abstentions. Crediting every abstention raises it to 836/1,221, still 2.99 points below DP-GRPO’s mean of 872.5 correct answers. This descriptive bound addresses unparsed base answers only, not errors in accepted extractions.